

Apalachicola National Estuarine Research Reserve

SEACAR Water Quality Analysis

Last compiled on 05 May, 2026

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Indicators

Nutrients

Total Nitrogen - Discrete

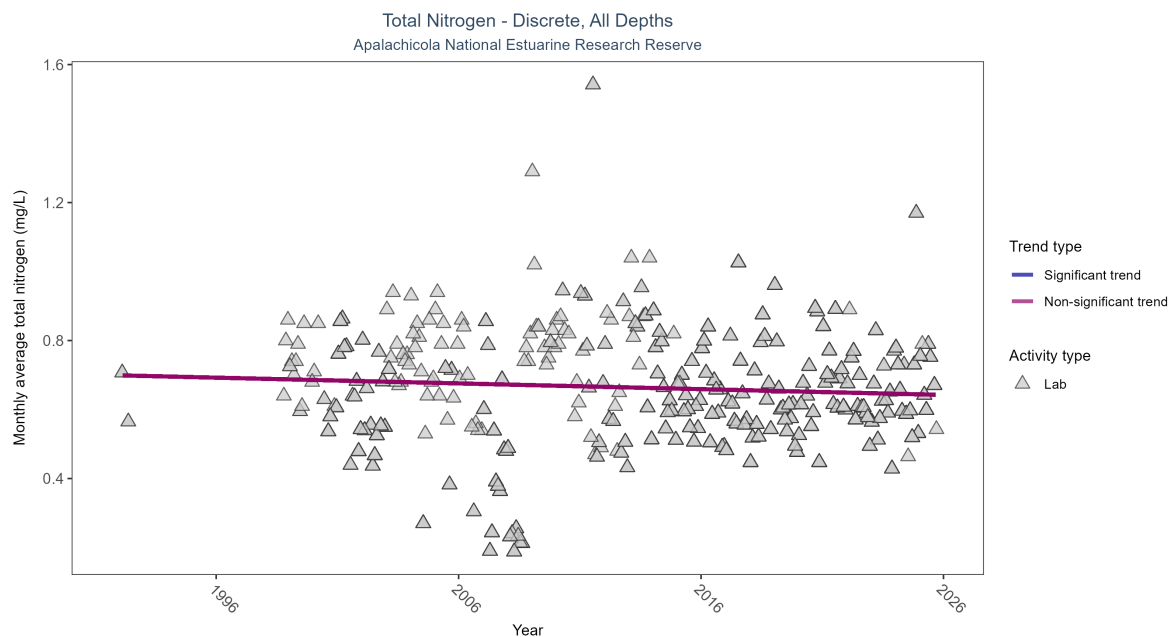


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	3831	29	1992 - 2025	0.628	-0.06347	0.69909	-0.00167	0.1357

Total nitrogen showed no detectable trend between 1992 and 2025.

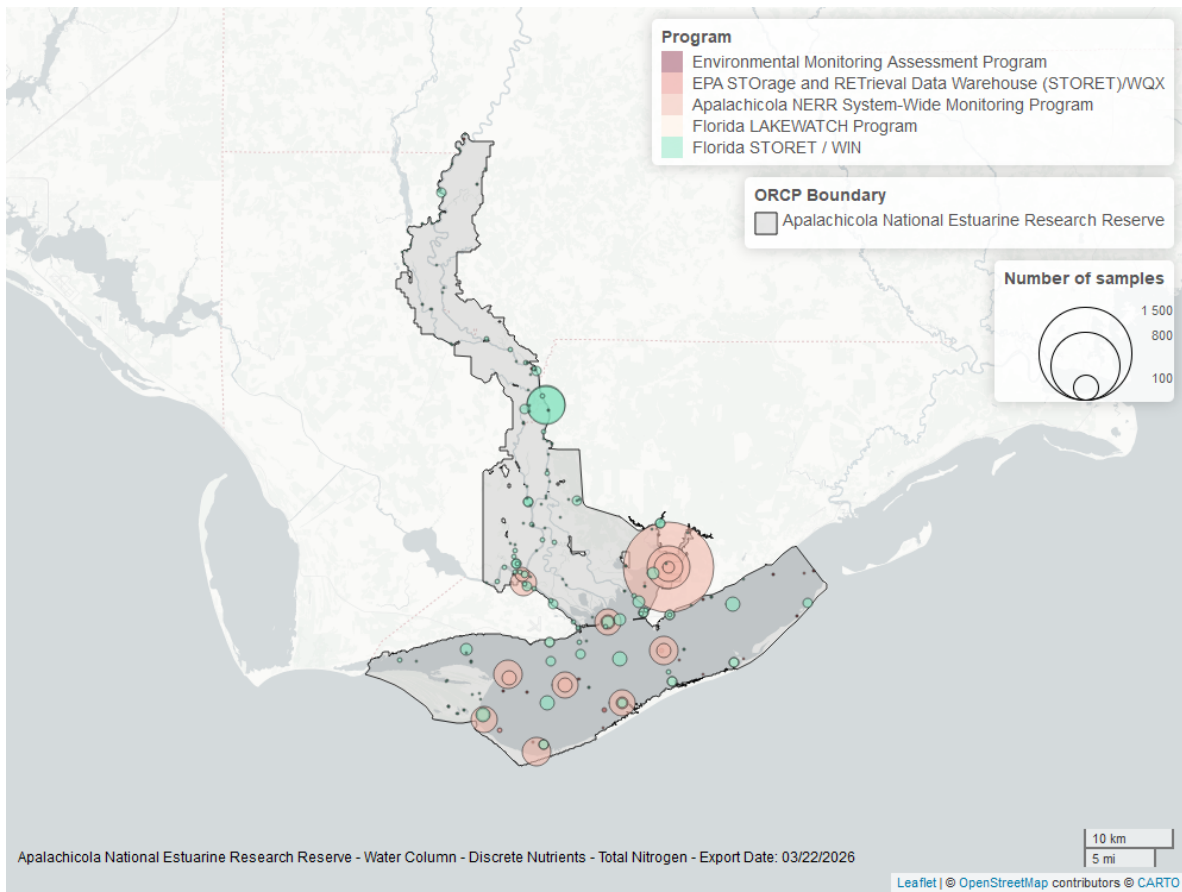


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Phosphorus - Discrete

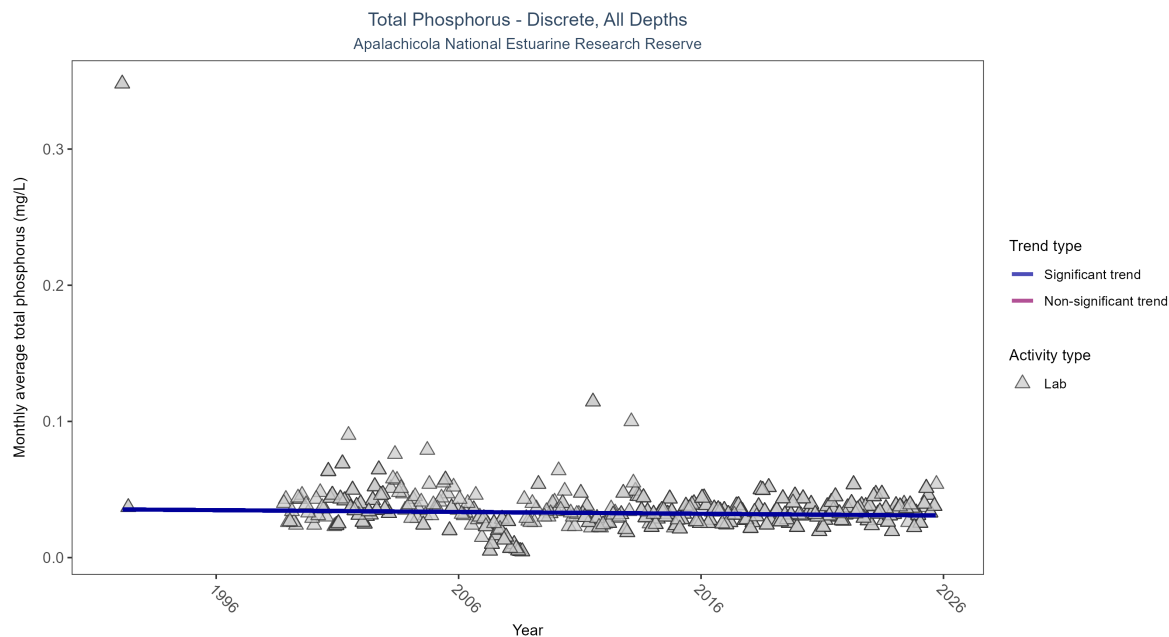


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	3970	29	1992 - 2025	0.031	-0.08501	0.03538	-0.00013	0.0345

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.

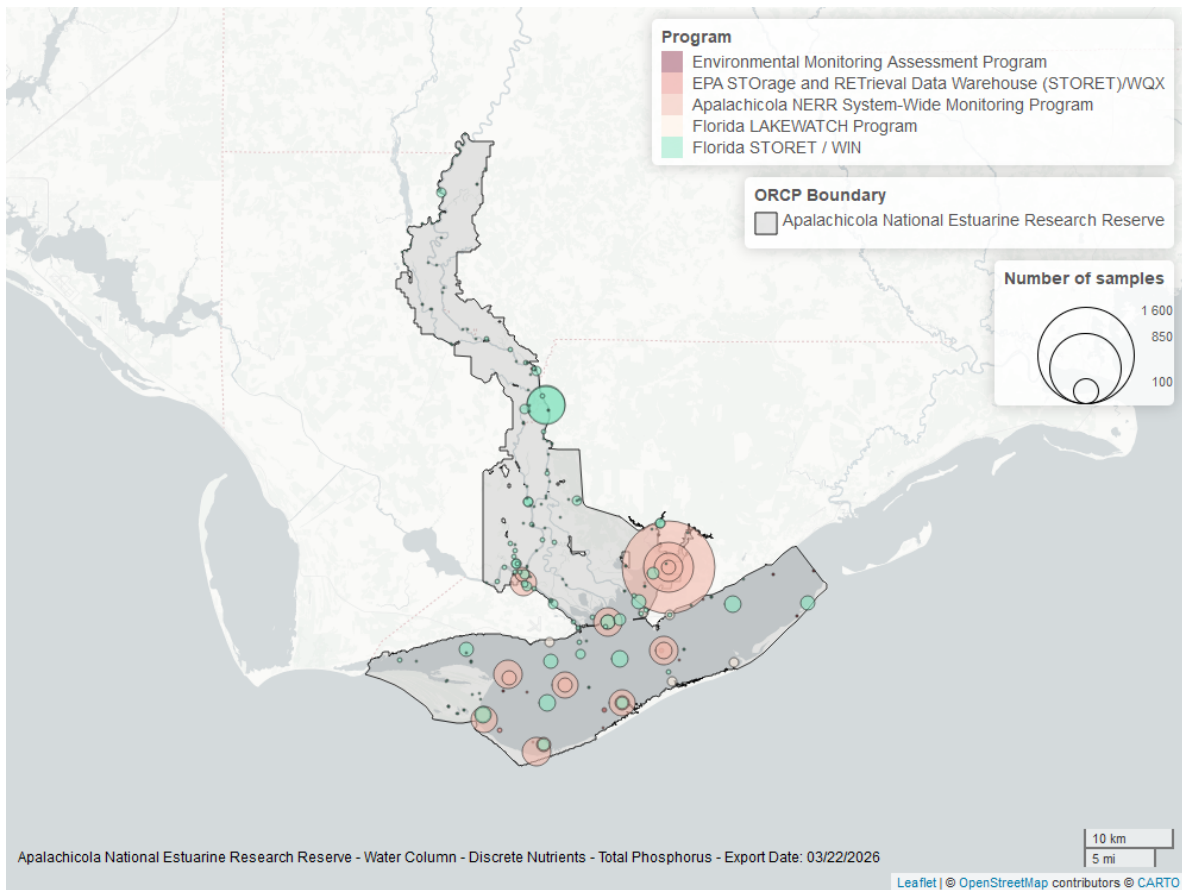


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Quality

Dissolved Oxygen - Continuous

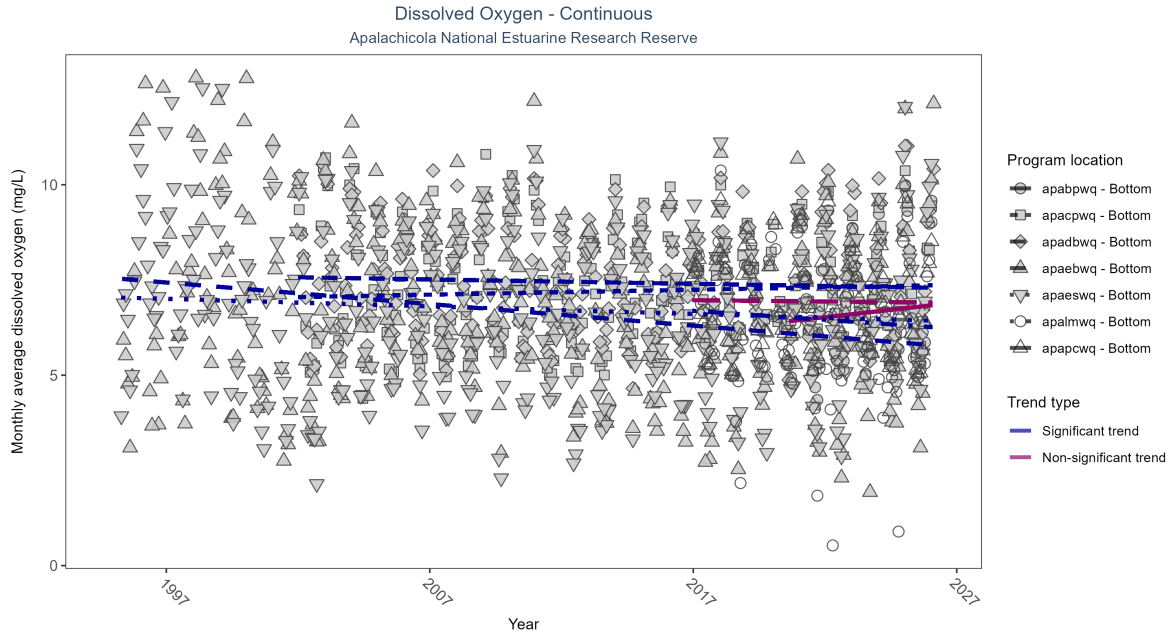


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Significantly decreasing trend	654044	25	2002 - 2026	7.3	-0.11	7.57	-0.01	0.0098
apaebwq	Significantly decreasing trend	698115	32	1995 - 2026	6.8	-0.27	7.55	-0.06	0
apaeswq	Significantly decreasing trend	745989	32	1995 - 2026	6.8	-0.10	7.04	-0.02	0.0067
apalmwq	Significantly decreasing trend	289063	11	2016 - 2026	6.3	-0.16	6.74	-0.05	0.0379
apapcwq	No significant trend	302205	11	2016 - 2026	6.9	-0.03	6.98	-0.01	0.7925
apabpwq	No significant trend	184169	7	2020 - 2026	6.5	0.14	6.37	0.08	0.1616
apacpwq	Significantly increasing trend	650006	25	2002 - 2026	7.1	0.12	7.05	0.01	0.0037

At one program location, monthly average dissolved oxygen increased by 0.01 mg/L per year. At four program locations, monthly average dissolved oxygen decreased between 0.01 and 0.06 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at two locations.

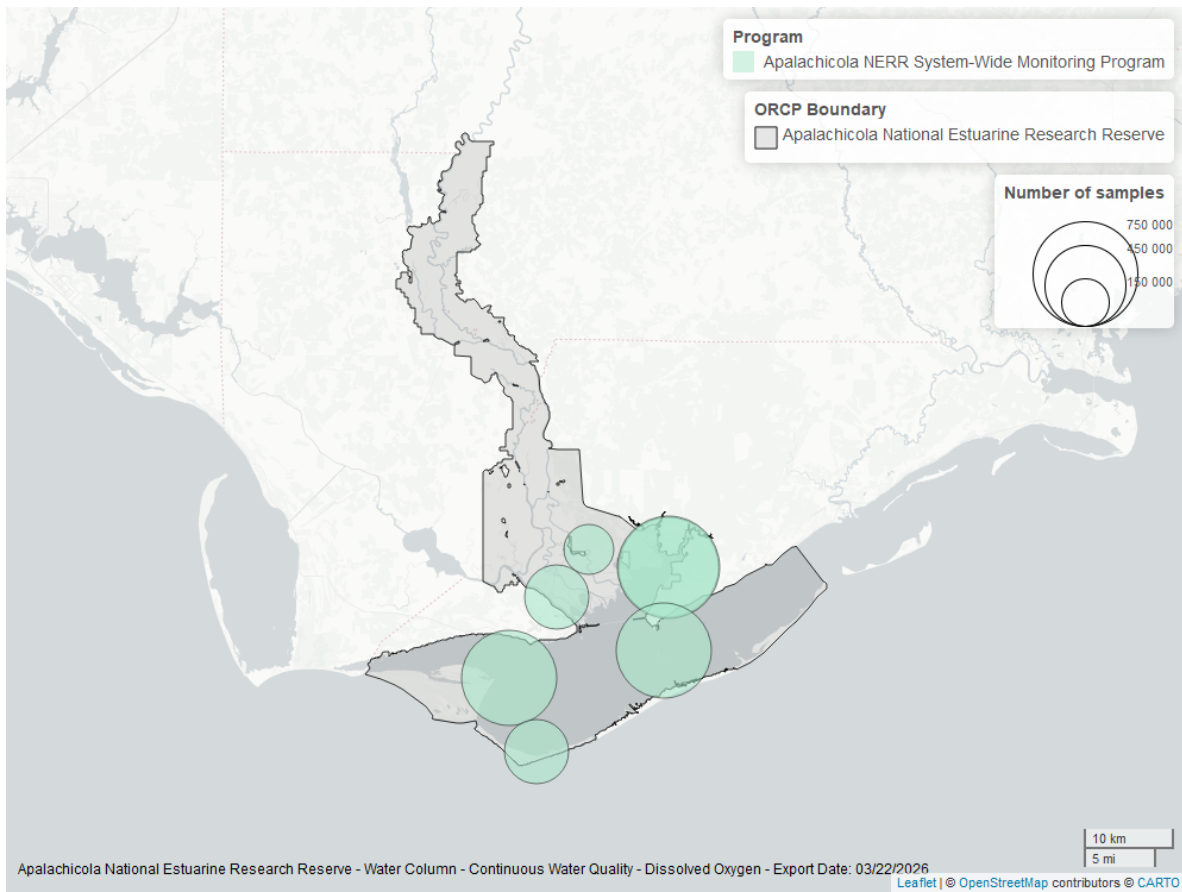


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen - Discrete

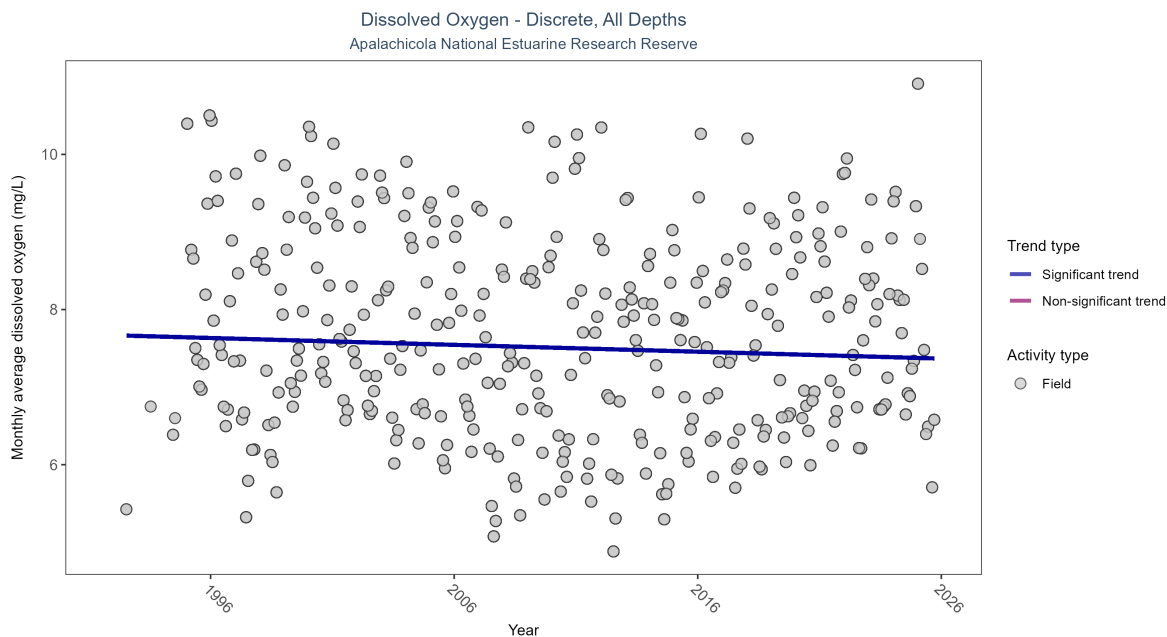


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	82148	34	1992 - 2025	7.5	-0.08287	7.66871	-0.00885	0.0252

Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

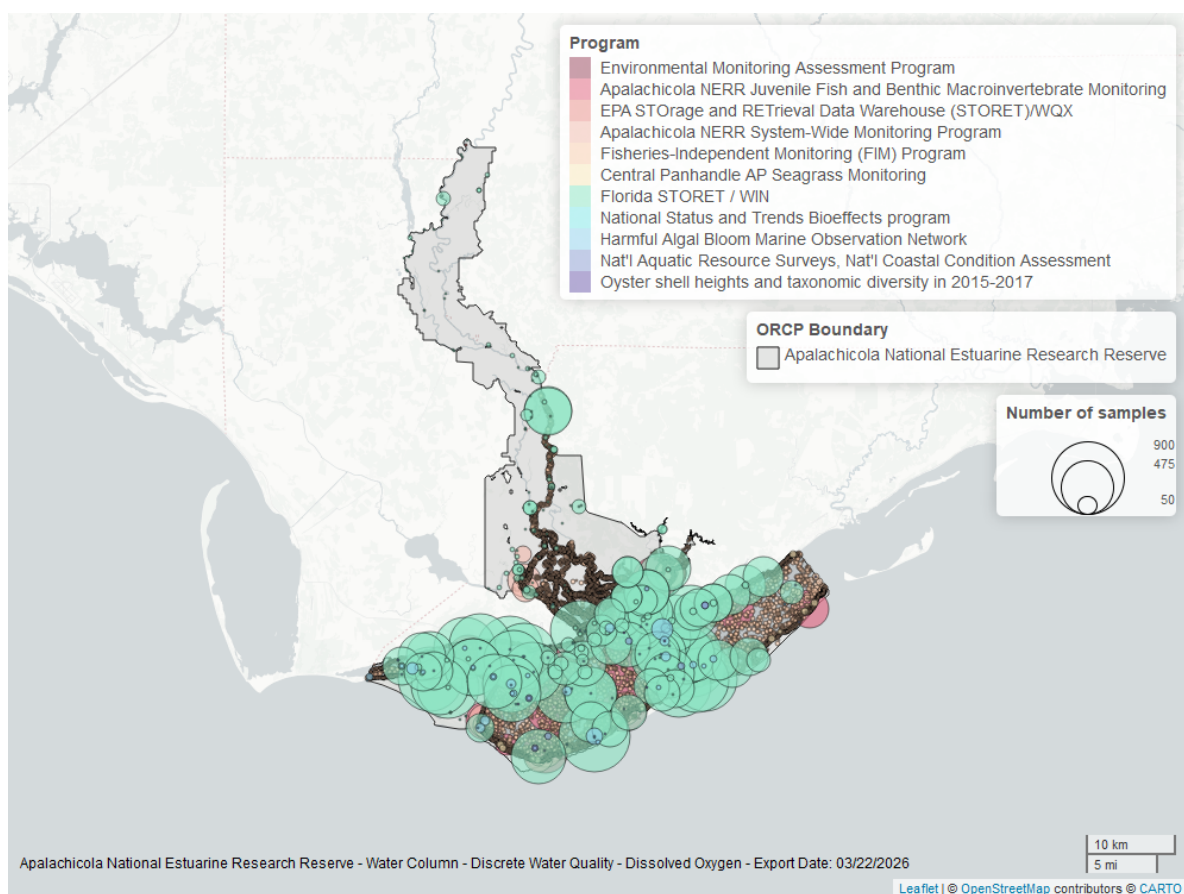


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Continuous

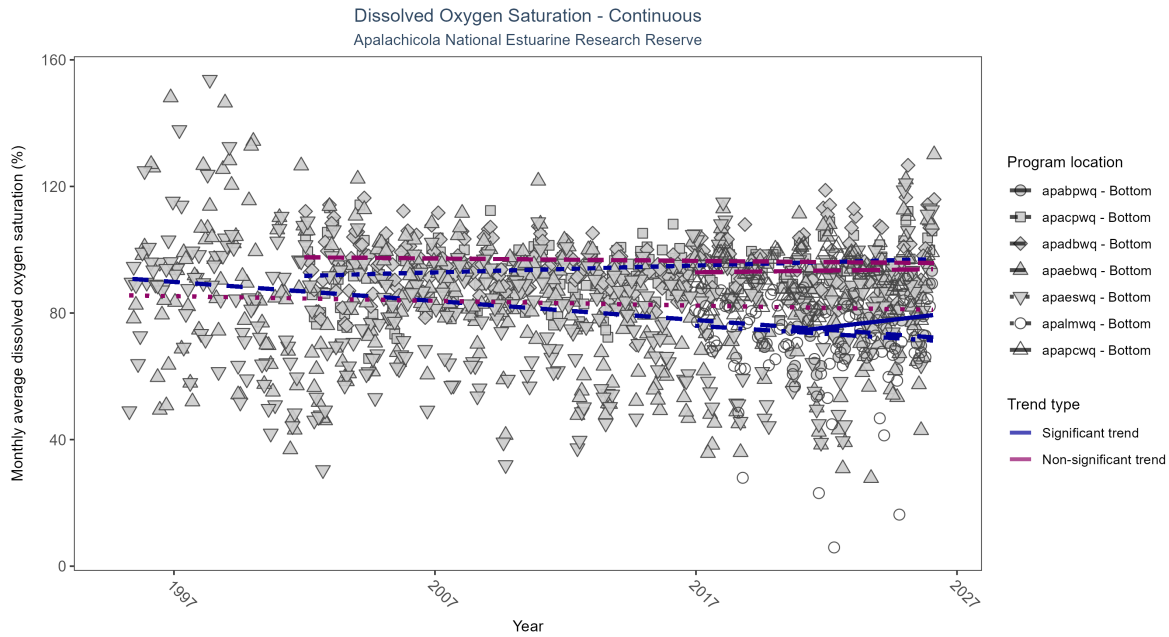


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No significant trend	658527	25	2002 - 2026	94.9	-0.05	97.62	-0.08	0.2718
apaebwq	Significantly decreasing trend	694367	32	1995 - 2026	84.9	-0.23	91.03	-0.60	0
apaeswq	No significant trend	746726	32	1995 - 2026	84.5	-0.06	85.63	-0.15	0.0976
apalmwq	Significantly decreasing trend	289599	11	2016 - 2026	74.9	-0.16	76.47	-0.51	0.0379
apapcwq	No significant trend	305661	11	2016 - 2026	94.0	0.04	92.74	0.11	0.5393
apabpwq	Significantly increasing trend	184169	7	2020 - 2026	77.0	0.25	73.51	0.96	0.0176
apacpwq	Significantly increasing trend	651534	25	2002 - 2026	94.3	0.18	91.75	0.22	0

At two program locations, monthly average dissolved oxygen saturation increased by 0.22% per year at one site and by 0.96% per year at the other. At two program locations, monthly average dissolved oxygen saturation decreased by 0.51% per year at one site and by 0.6% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at three locations.

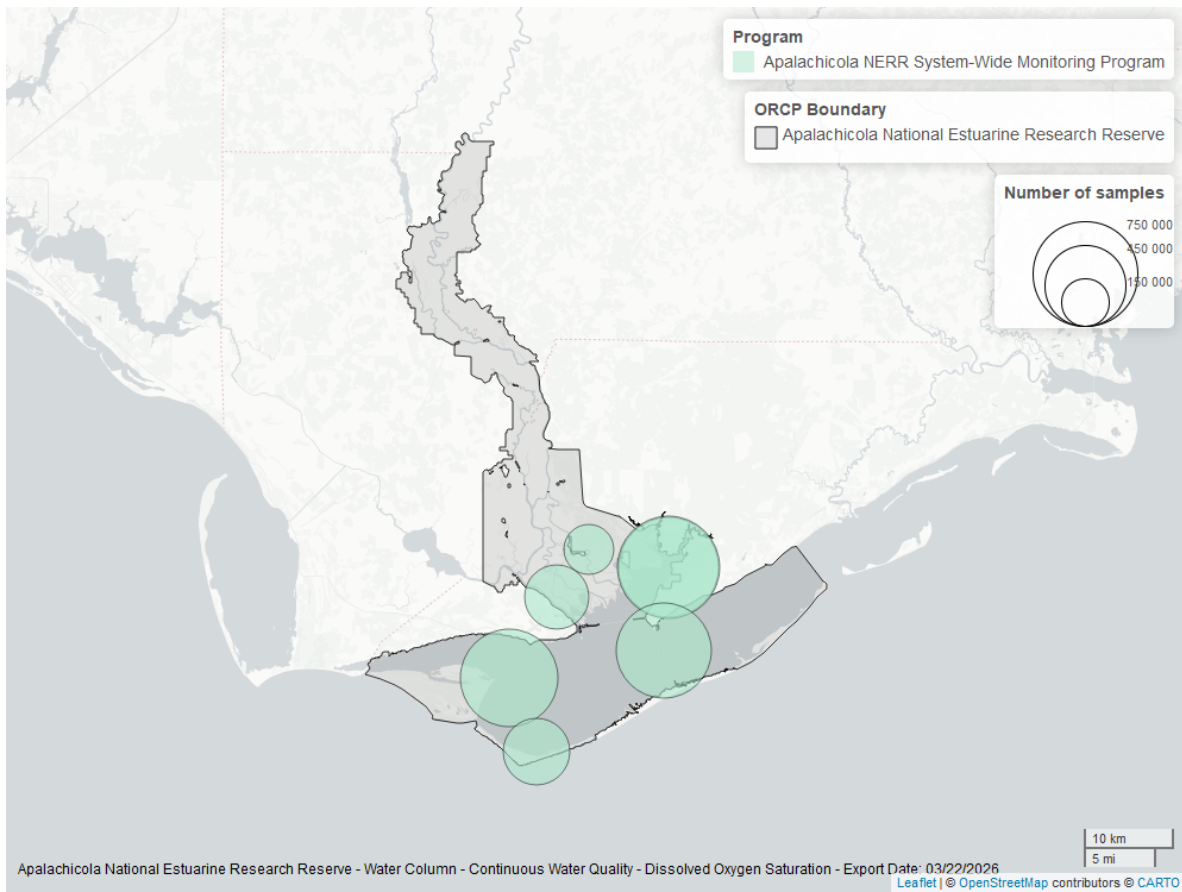


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Dissolved Oxygen Saturation - Discrete

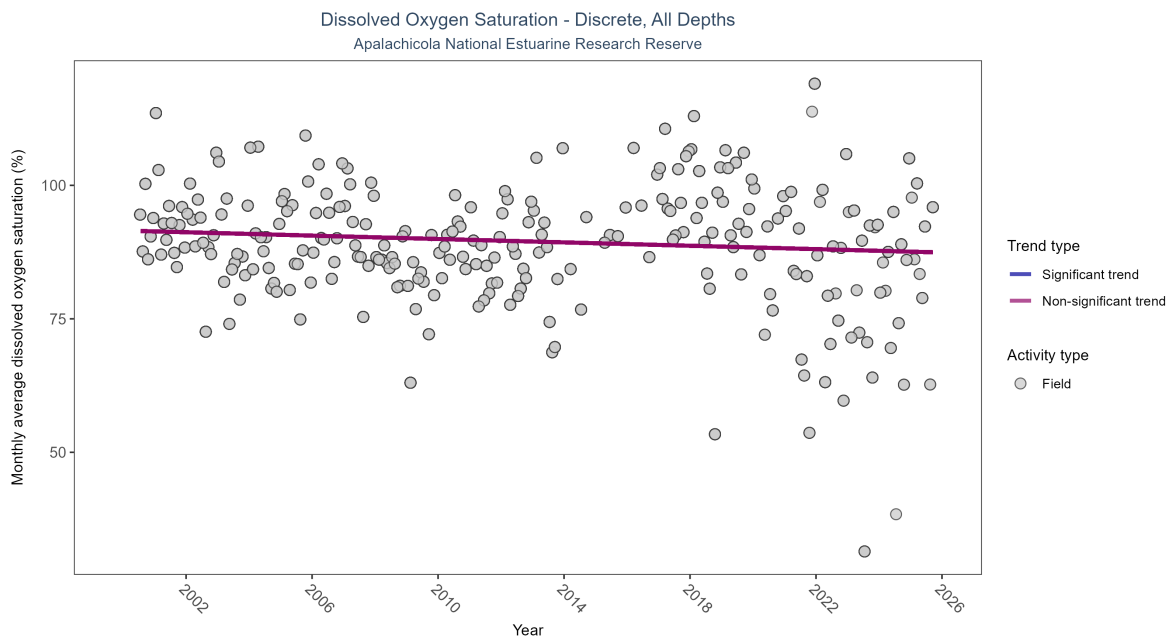


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No significant trend	5833	26	2000 - 2025	91.4	-0.09071	91.53316	-0.15814	0.0515

Dissolved oxygen saturation showed no detectable trend between 2000 and 2025.

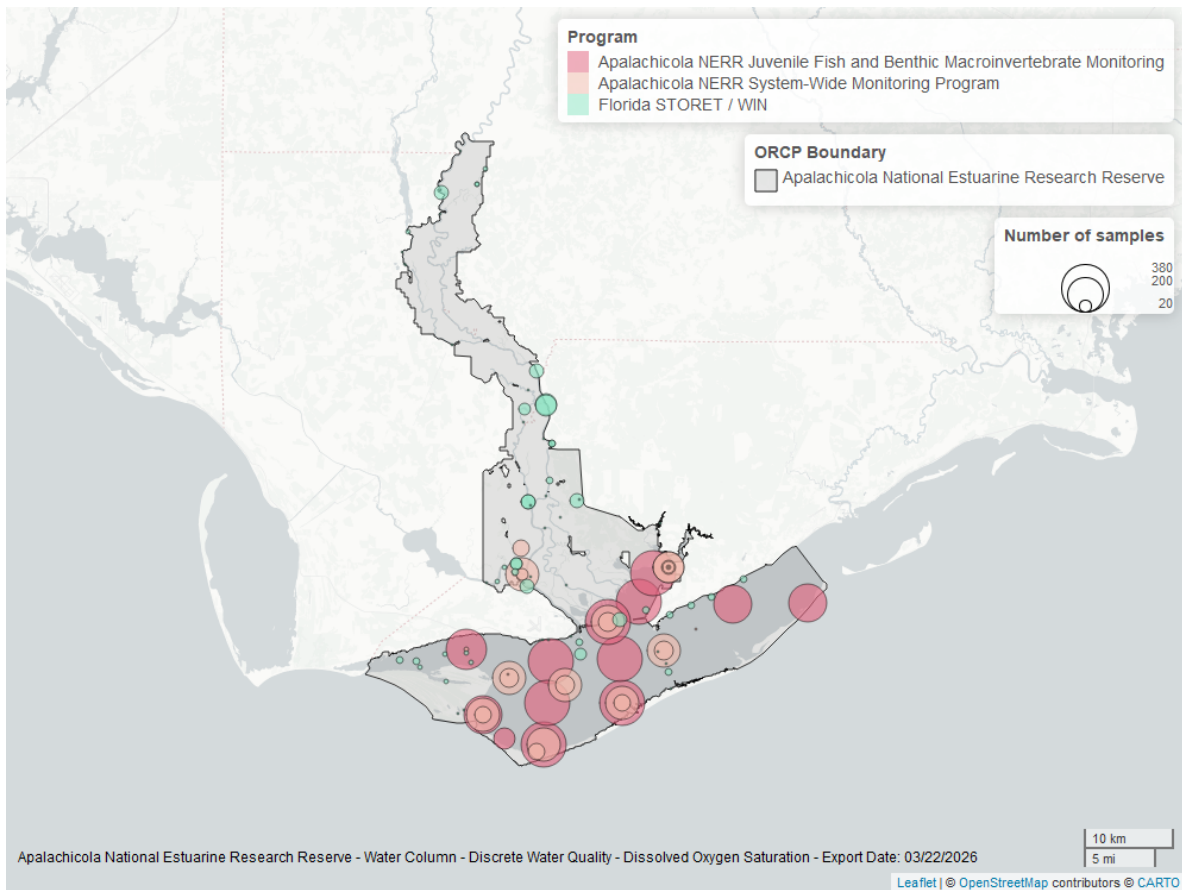


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Discrete

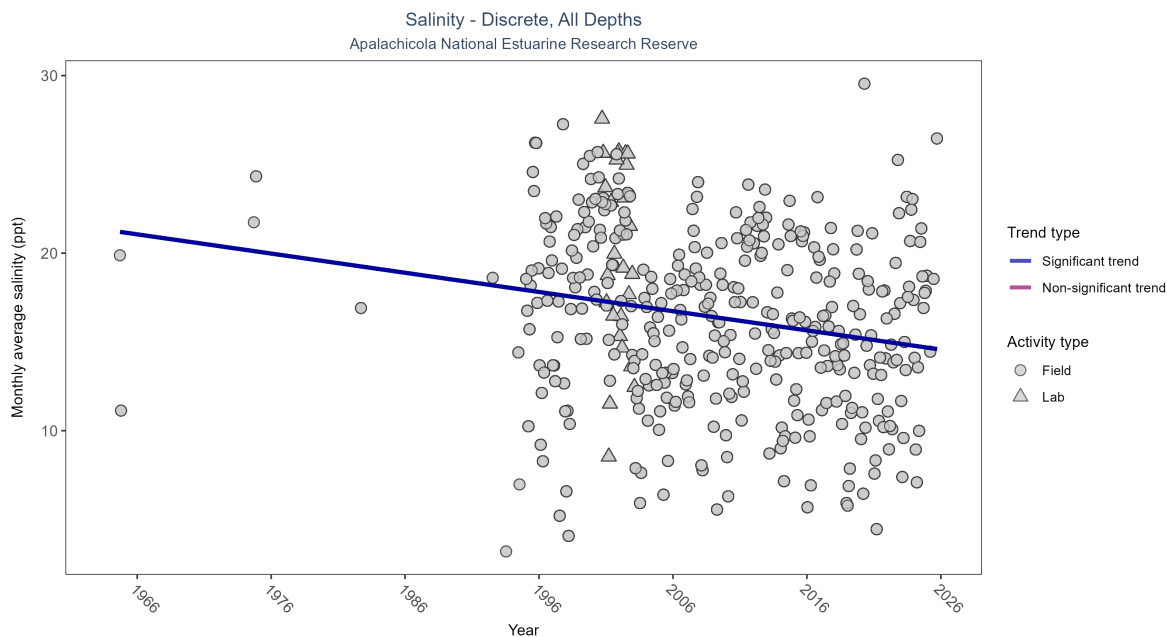


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	Significantly decreasing trend	96103	37	1964 - 2025	16.8	-0.16808	21.27605	-0.10819	0

Monthly average salinity decreased by 0.11 ppt per year.

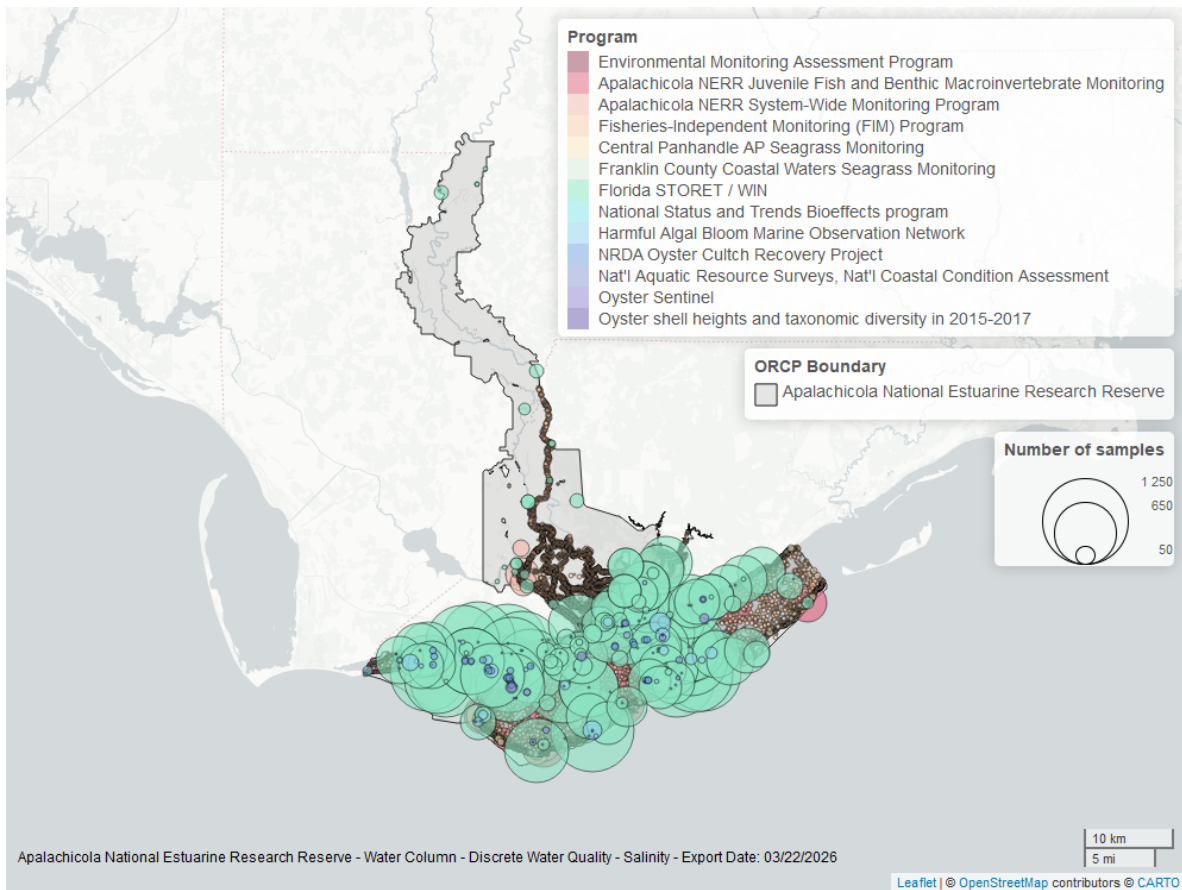


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous



Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No significant trend	651798	25	2002 - 2026	22.3	0.00	21.84	-0.01	0.9332
apaebwq	No significant trend	782974	32	1995 - 2026	10.0	0.04	9.24	0.02	0.3201
apaeswq	No significant trend	792598	32	1995 - 2026	7.6	0.05	6.97	0.03	0.1789
apalmwq	Significantly increasing trend	304777	11	2016 - 2026	0.1	0.24	0.08	0.01	0.0014
apapcwq	No significant trend	303665	11	2016 - 2026	27.1	0.14	25.99	0.25	0.0655
apabpwq	Significantly increasing trend	186721	7	2020 - 2026	0.1	0.34	0.06	0.01	0.001
apacpwq	No significant trend	666150	25	2002 - 2026	22.5	0.01	21.69	0.01	0.8105

At two program locations, monthly average salinity increased by 0.01 ppt per year. No detectable change in monthly average salinity was observed at five locations.

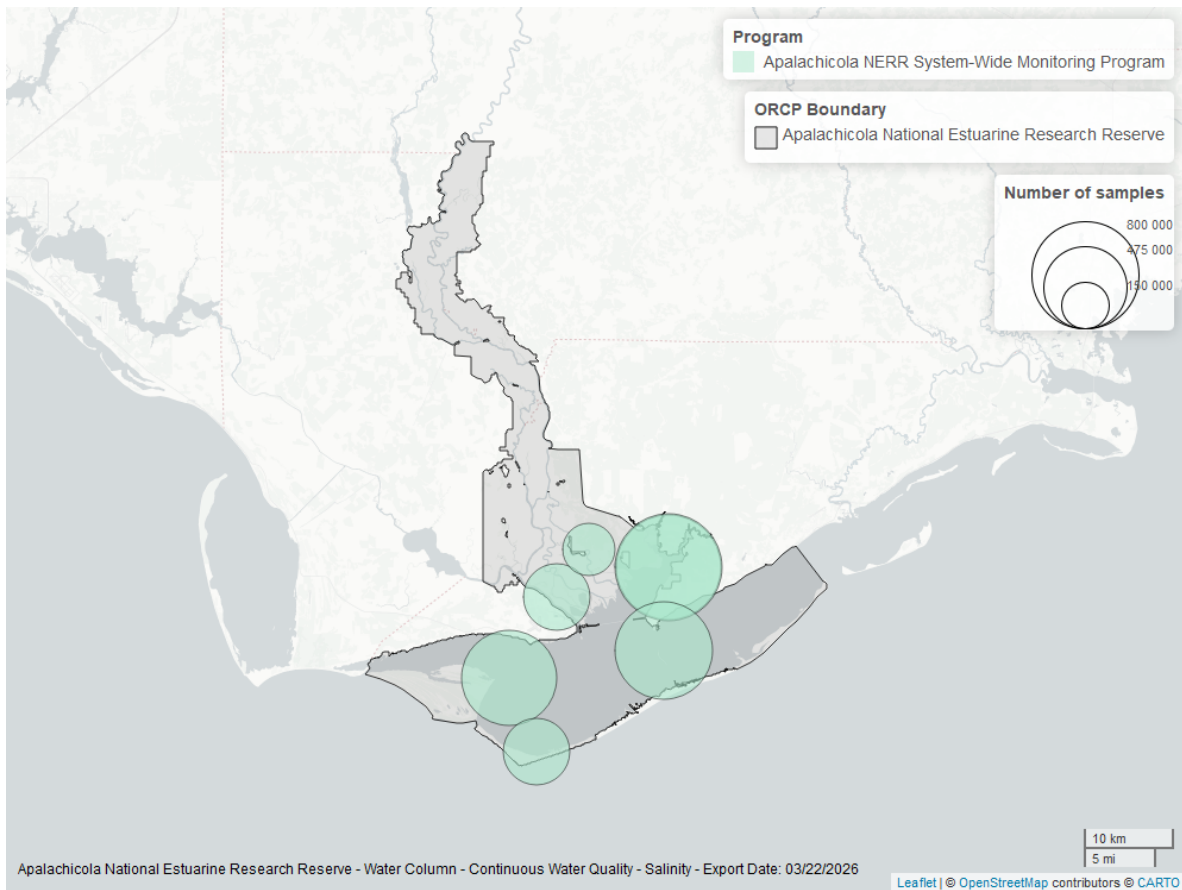


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Continuous

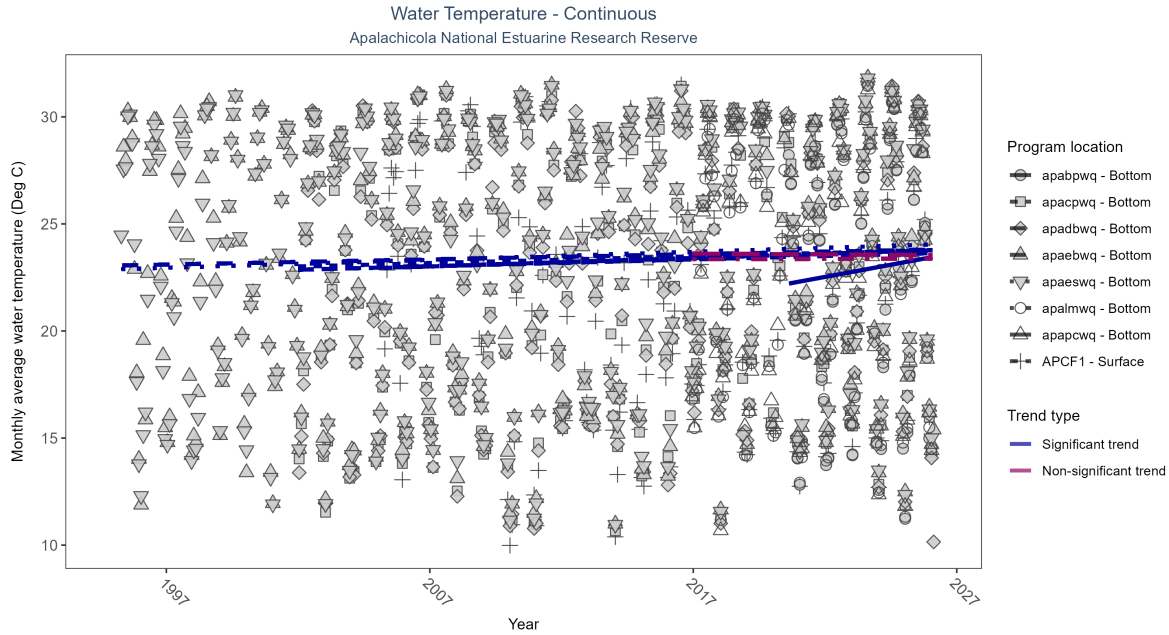


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Significantly increasing trend	679646	25	2002 - 2026	23.3	0.16	22.86	0.03	2e-04
apaebwq	Significantly increasing trend	799283	32	1995 - 2026	24.1	0.16	23.05	0.02	0
apaeswq	Significantly increasing trend	800593	32	1995 - 2026	24.1	0.20	22.88	0.04	0
apalmwq	No significant trend	307034	11	2016 - 2026	22.7	0.01	23.35	0.00	0.9301
apapcwq	No significant trend	308627	11	2016 - 2026	23.2	0.00	23.62	-0.01	0.9767
APCF1	Significantly increasing trend	1433809	21	2005 - 2025	23.4	0.12	22.93	0.04	0.0181
apabpwq	Significantly increasing trend	186721	7	2020 - 2026	22.6	0.23	22.07	0.23	0.0446
apacpwq	Significantly increasing trend	705144	25	2002 - 2026	23.5	0.16	23.01	0.03	1e-04

At six program locations, monthly average water temperature increased between 0.02 and 0.23°C per year. No detectable change in monthly average water temperature was observed at two locations.

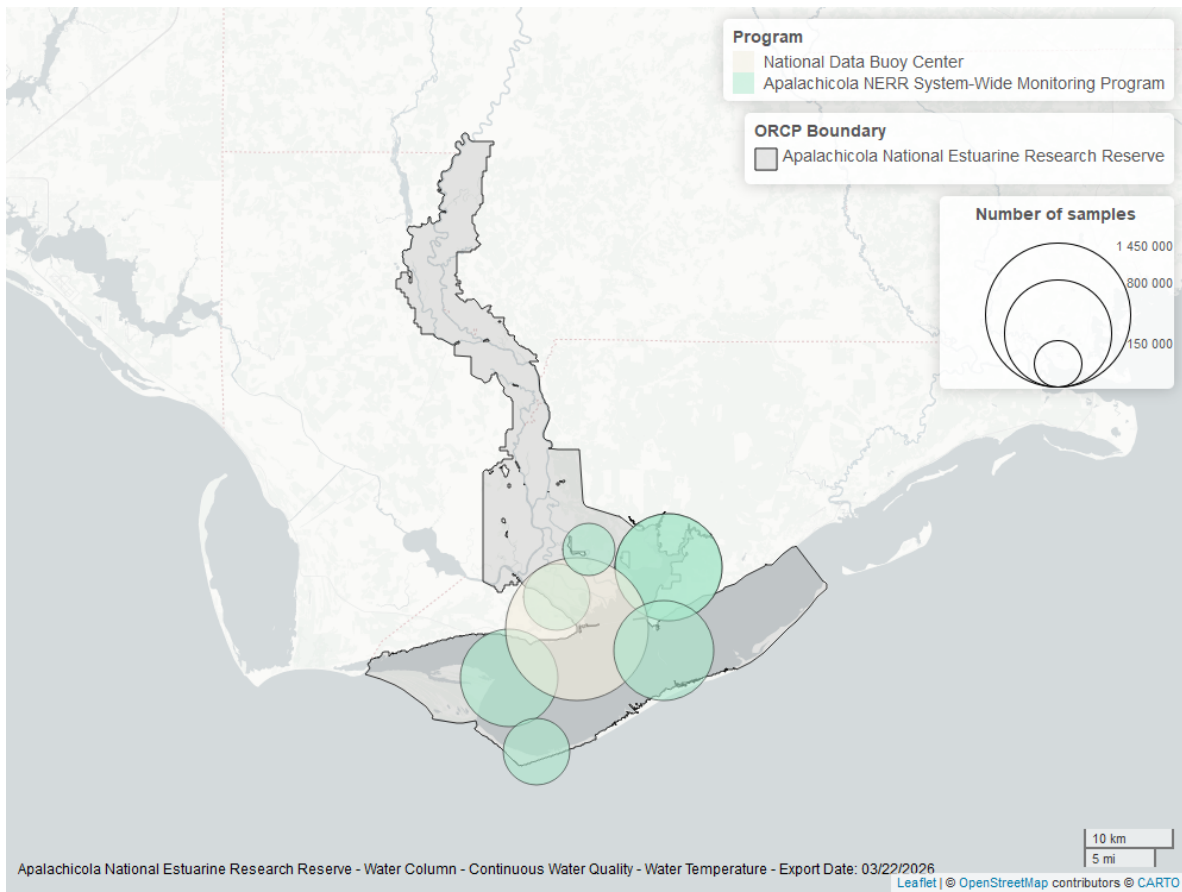


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Temperature - Discrete

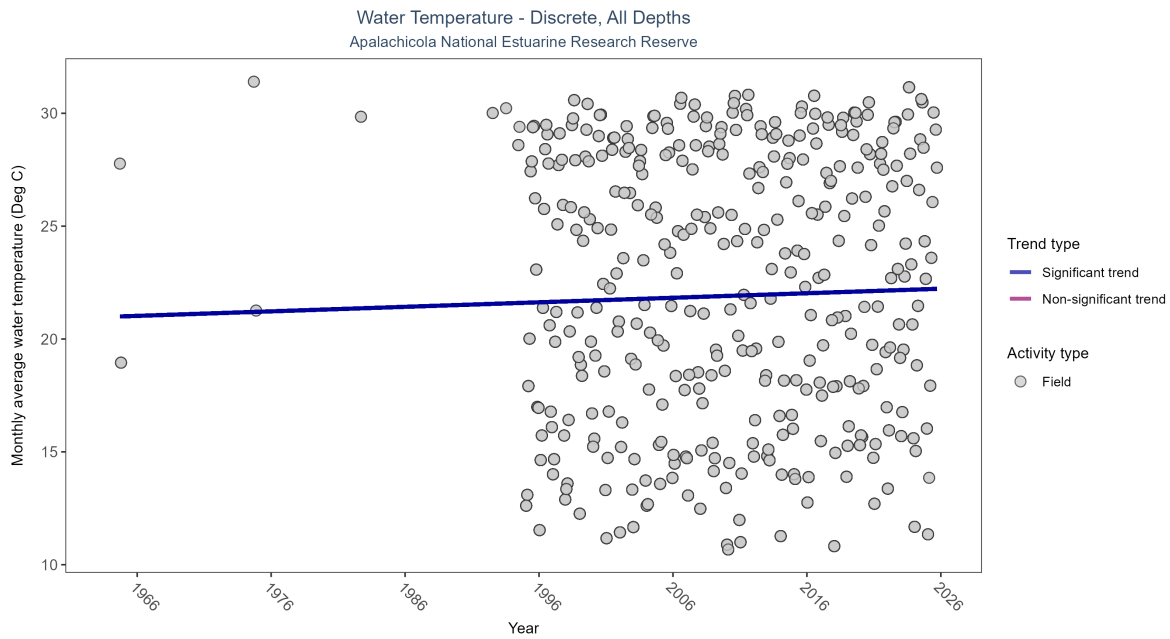


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	95140	37	1964 - 2025	23.4	0.12621	20.98404	0.02006	5e-04

Monthly average water temperature increased by 0.02°C per year.

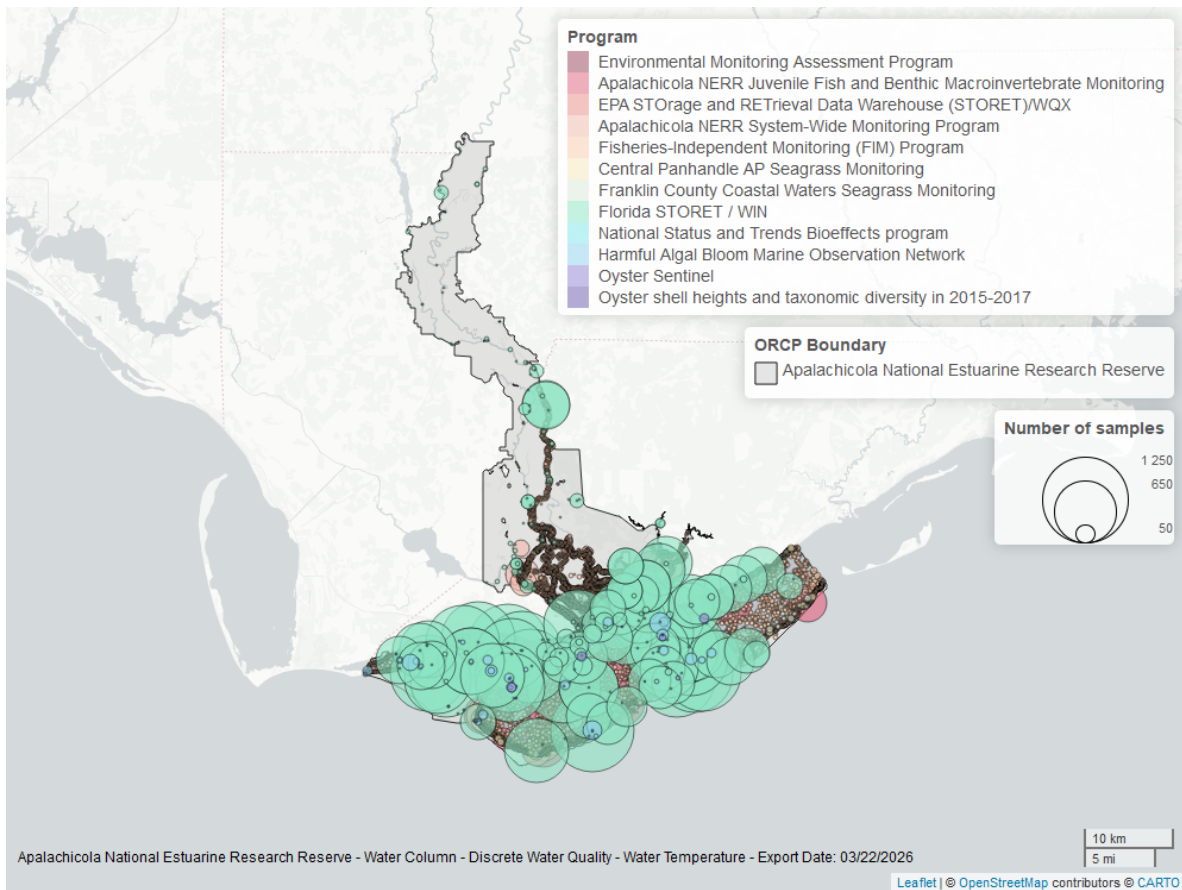


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Continuous

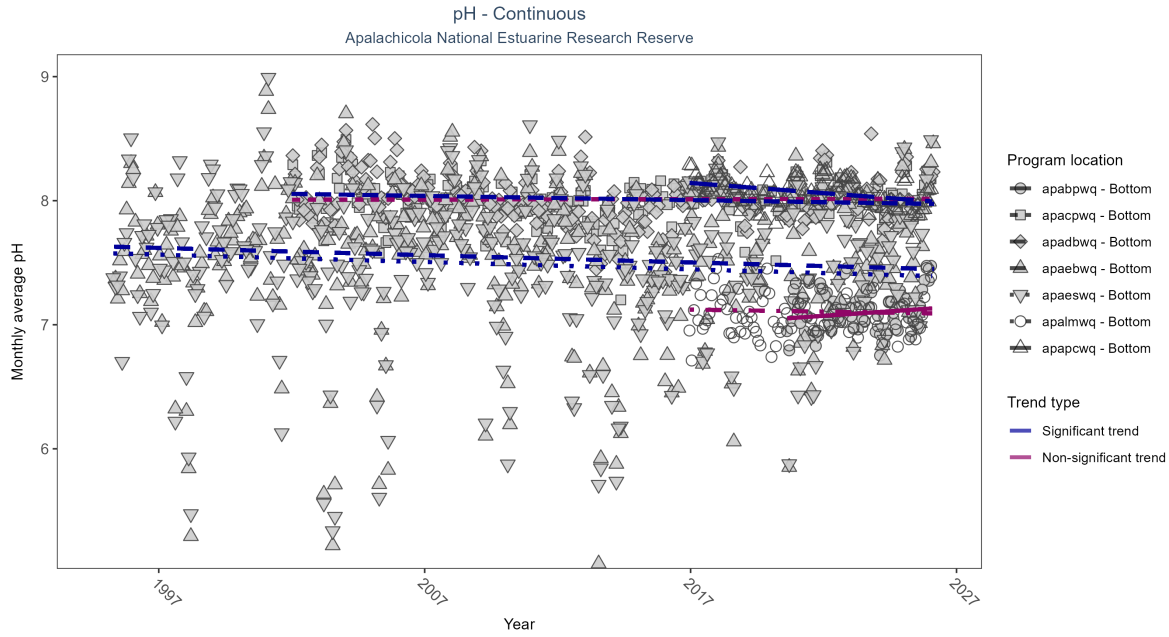


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	Significantly decreasing trend	636788	25	2002 - 2026	8.0	-0.11	8.06	0.00	0.0082
apaebwq	Significantly decreasing trend	754923	32	1995 - 2026	7.6	-0.11	7.63	-0.01	0.0038
apaeswq	Significantly decreasing trend	745836	32	1995 - 2026	7.5	-0.10	7.58	-0.01	0.0081
apalmwq	No significant trend	297196	11	2016 - 2026	7.1	-0.06	7.13	0.00	0.5013
apapcwq	Significantly decreasing trend	302765	11	2016 - 2026	8.1	-0.35	8.16	-0.02	0
apabpwq	No significant trend	184904	7	2020 - 2026	7.1	0.09	7.04	0.01	0.3009
apacpwq	No significant trend	649552	25	2002 - 2026	8.0	0.01	8.01	0.00	0.8559

At four program locations, monthly average pH decreased between less than 0.01 and 0.02 pH units per year. No detectable change in monthly average pH was observed at three locations.

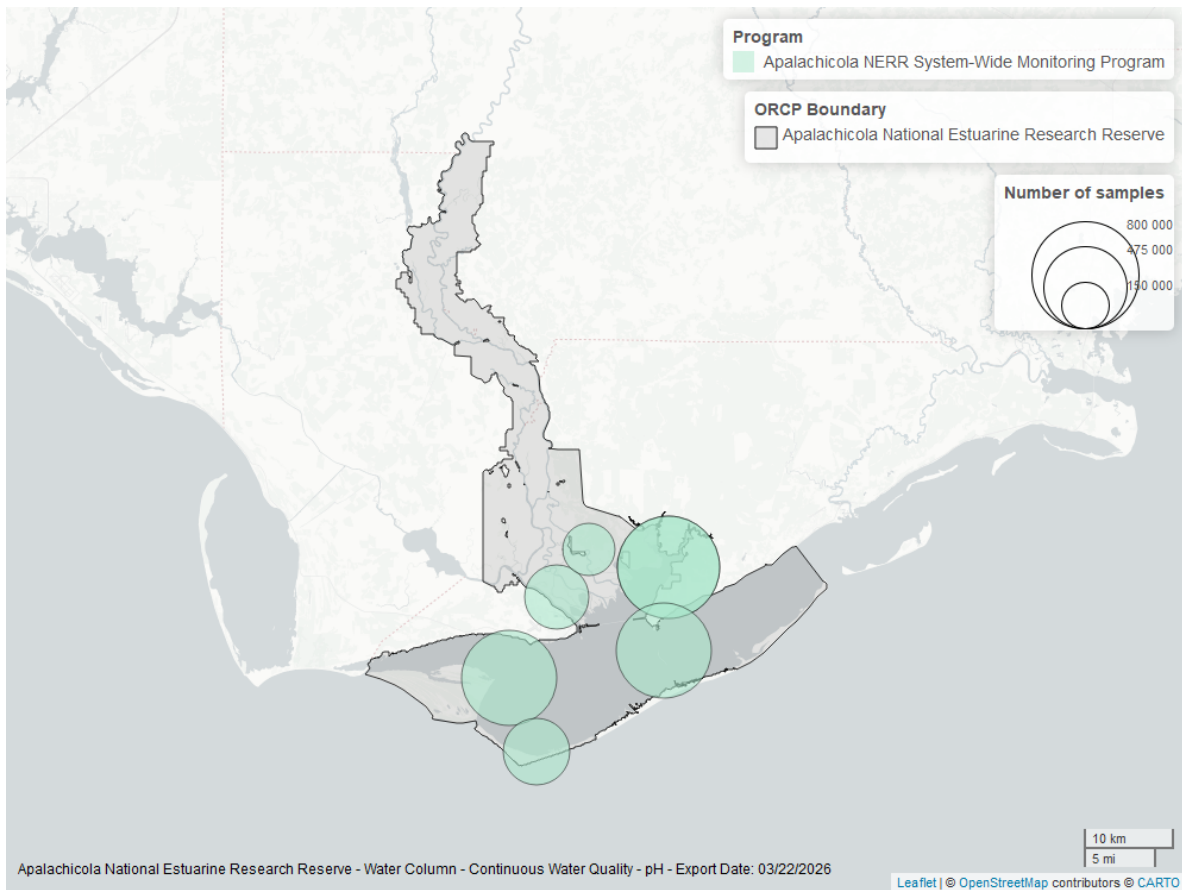


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

pH - Discrete

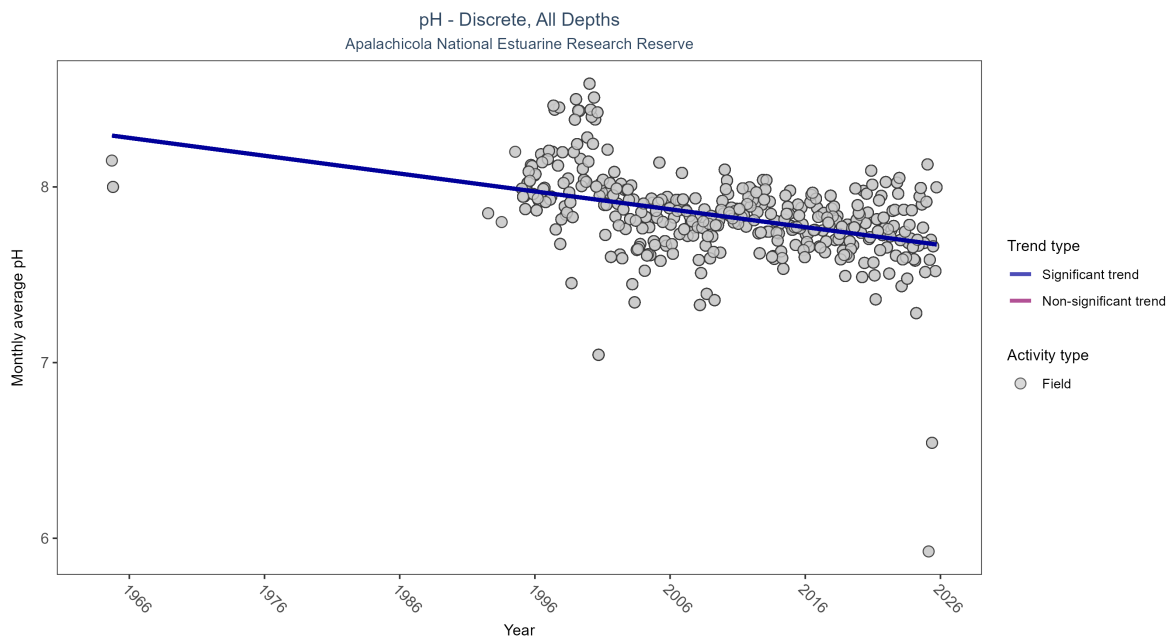


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly decreasing trend	67283	35	1964 - 2025	7.98	-0.34086	8.29946	-0.01016	0

Monthly average pH decreased by 0.01 pH units per year.

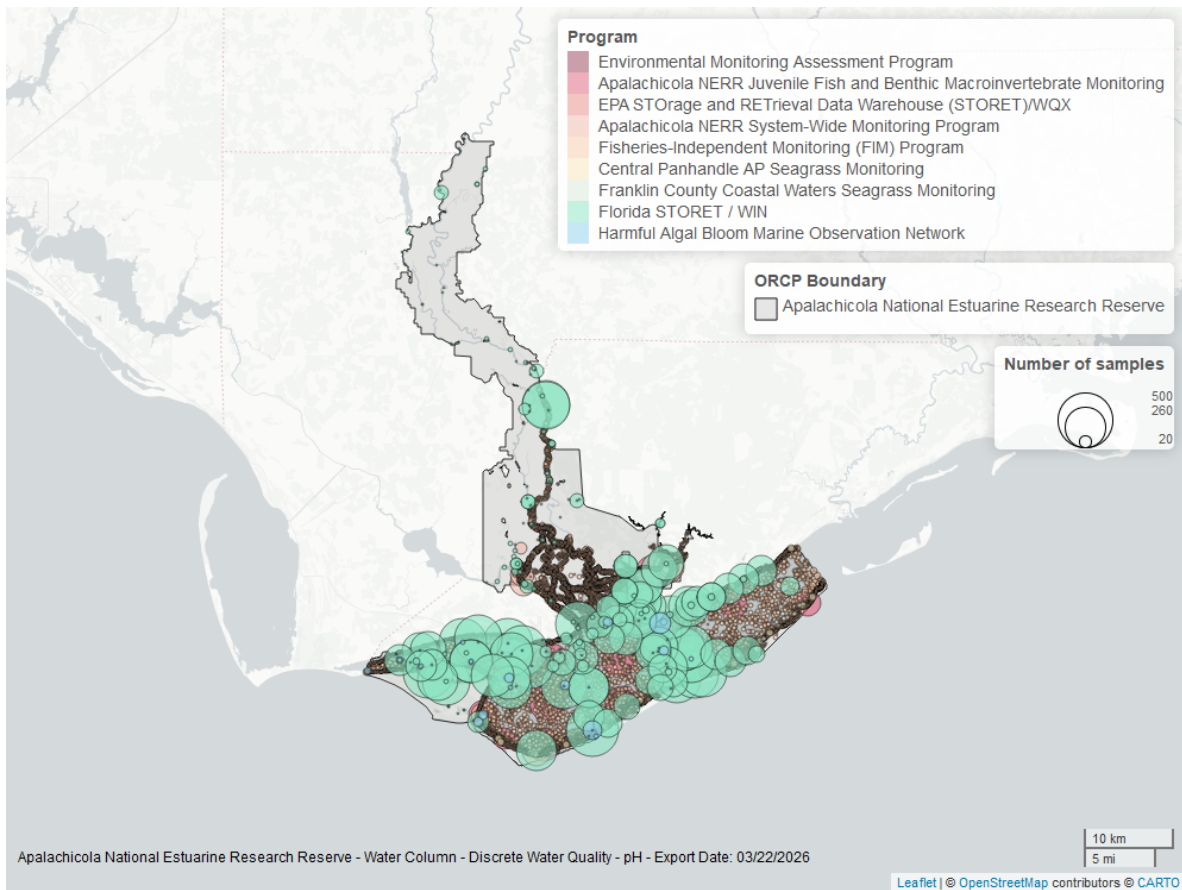


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

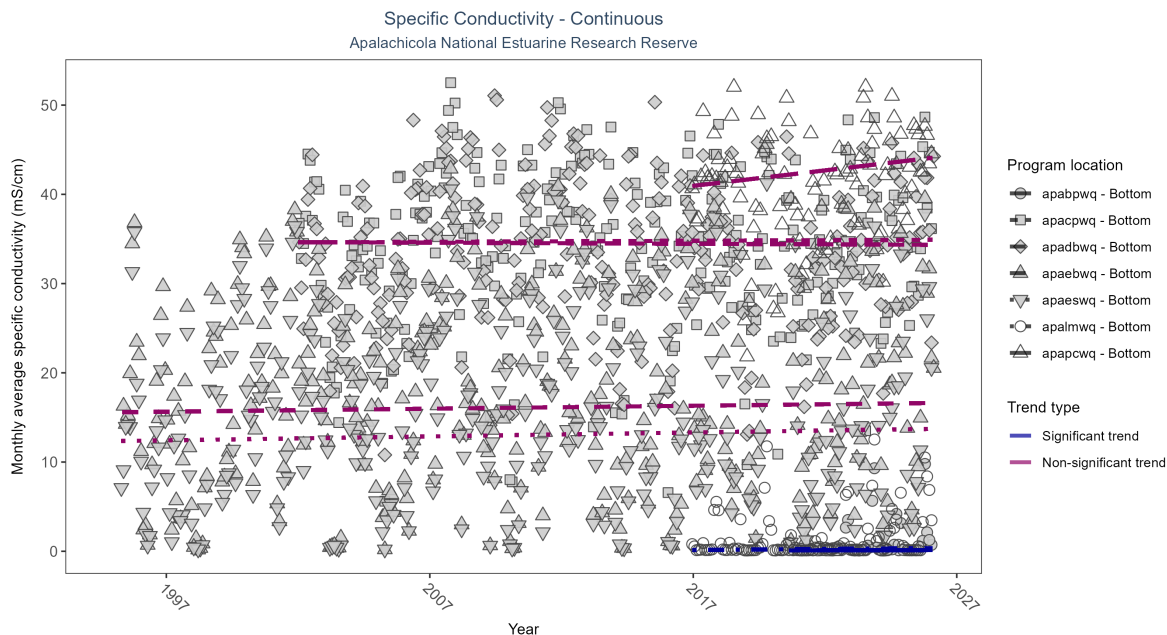


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbqw	No significant trend	651297	25	2002 - 2026	35.35	-0.01	34.64	-0.01	0.8609
apaebwq	No significant trend	783182	32	1995 - 2026	16.97	0.04	15.57	0.03	0.3201
apaeswq	No significant trend	792928	32	1995 - 2026	13.14	0.05	12.34	0.04	0.2
apalmwq	Significantly increasing trend	304777	11	2016 - 2026	0.15	0.24	0.12	0.02	0.0018
apapcwq	No significant trend	303667	11	2016 - 2026	42.28	0.15	40.60	0.35	0.0574
apabpwq	Significantly increasing trend	186713	7	2020 - 2026	0.12	0.27	0.10	0.01	0.0089
apacpwq	No significant trend	666166	25	2002 - 2026	35.70	0.01	34.62	0.01	0.8673

At two program locations, monthly average specific conductivity increased by 0.01 ms/cm per year at one site and by 0.02 ms/cm per year at the other. No detectable change in monthly average specific conductivity was observed at five locations.

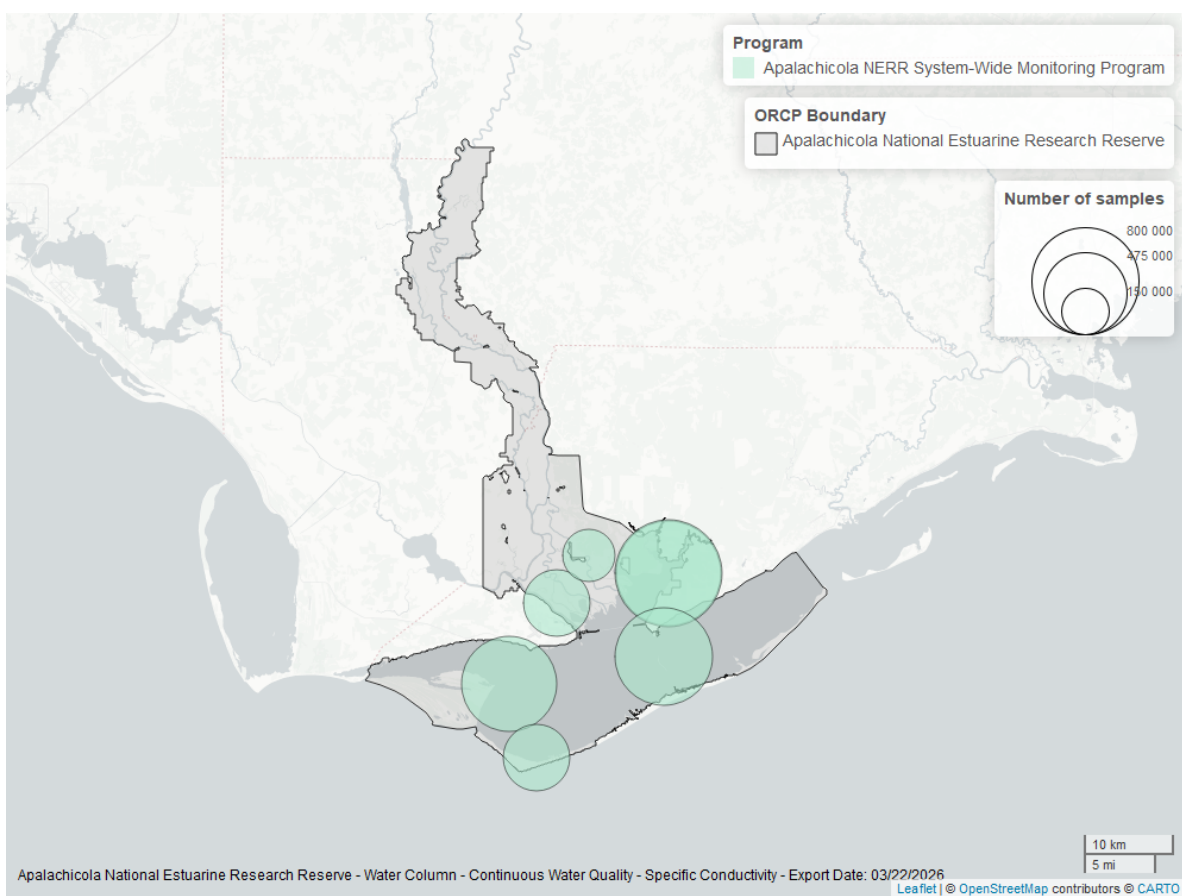


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Water Clarity

Turbidity - Continuous

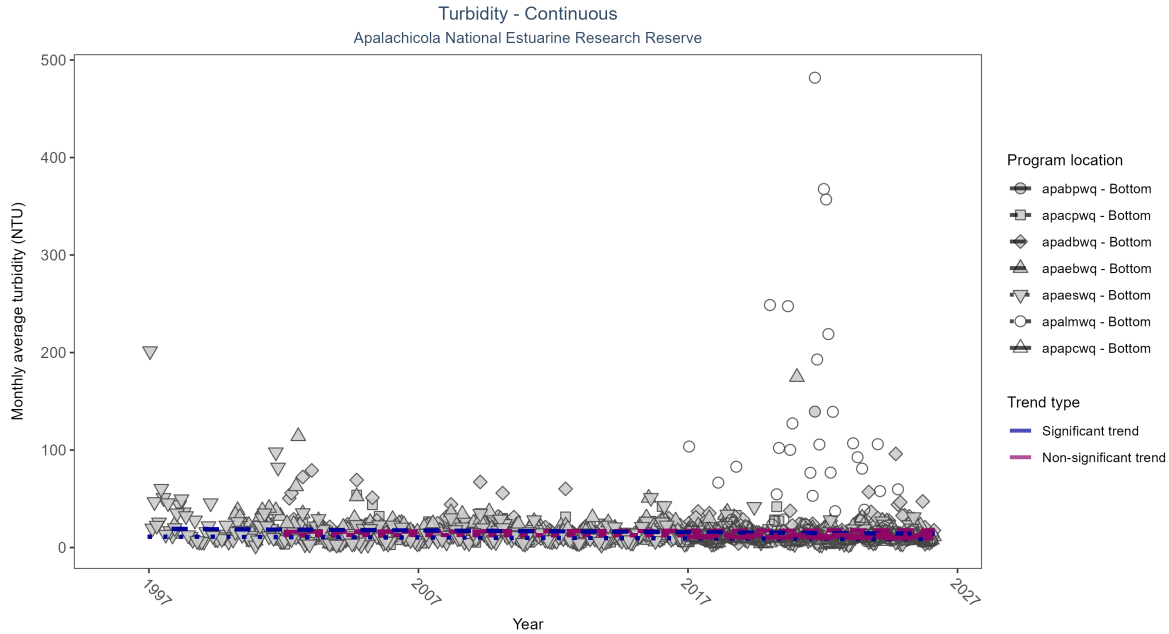


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

Program Location	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
apadbwq	No significant trend	638719	25	2002 - 2026	10	0.05	16.21	0.06	0.2341
apaebwq	Significantly decreasing trend	672490	28	1997 - 2026	13	-0.18	19.33	-0.18	0
apaeswq	Significantly decreasing trend	735494	31	1996 - 2026	9	-0.13	11.31	-0.09	0.001
apalmwq	No significant trend	275282	11	2016 - 2026	11	0.08	11.53	0.23	0.3592
apapcwq	No significant trend	292616	11	2016 - 2026	7	-0.11	10.95	-0.15	0.1473
apabpwq	No significant trend	186715	7	2020 - 2026	11	-0.07	11.82	-0.20	0.4289
apacpwq	No significant trend	663199	25	2002 - 2026	8	-0.02	12.90	-0.02	0.5896

At two program locations, monthly average turbidity decreased by 0.09 NTU per year at one site and by 0.18 NTU per year at the other. No detectable change in monthly average turbidity was observed at five locations.

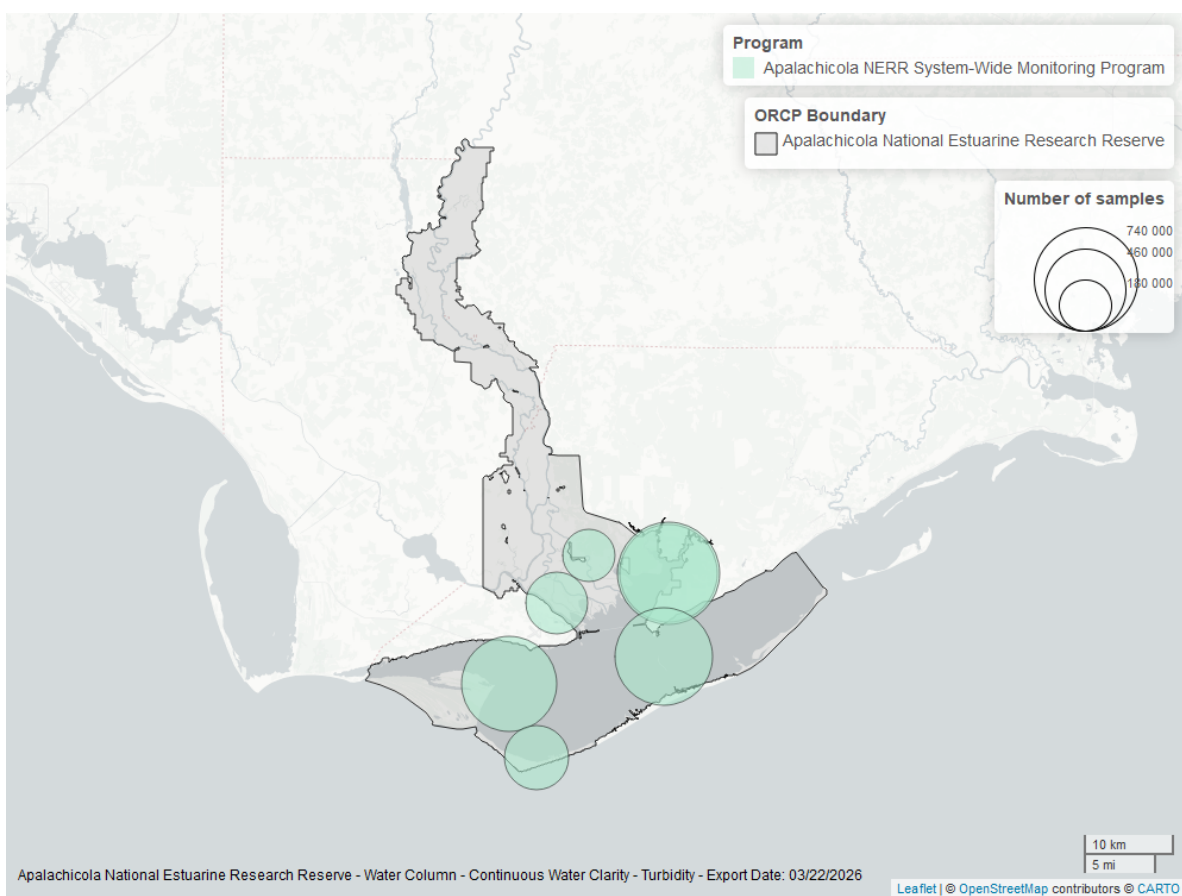


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Turbidity - Discrete

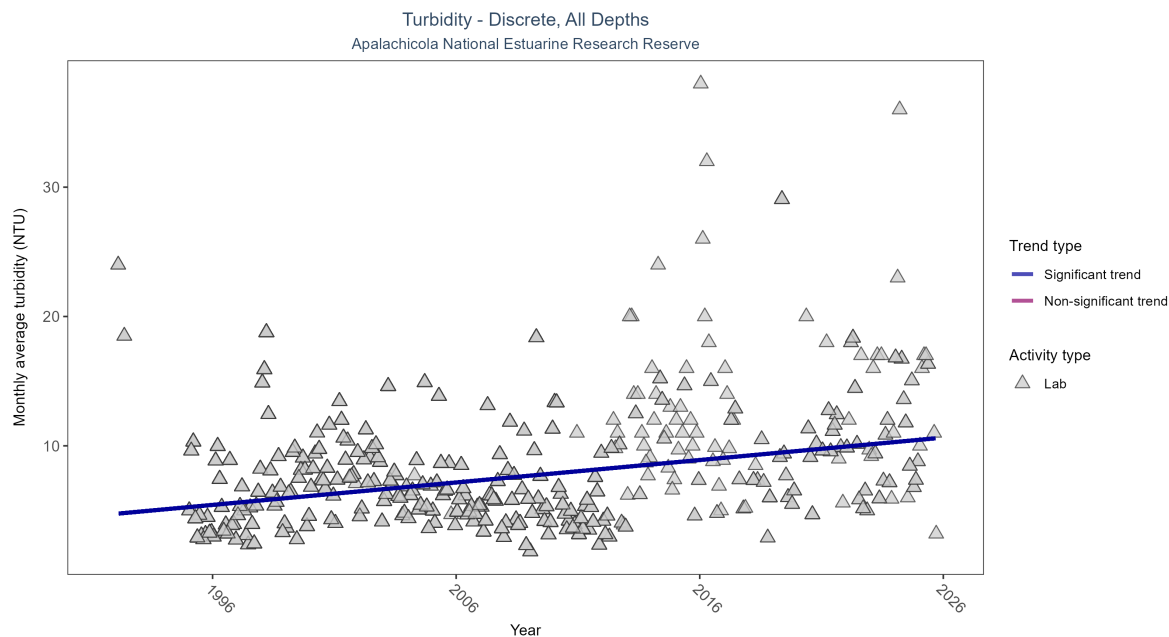


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	22996	32	1992 - 2025	5.1	0.27472	4.73754	0.17348	0

Monthly average turbidity increased by 0.17 NTU per year, indicating a decrease in water clarity.

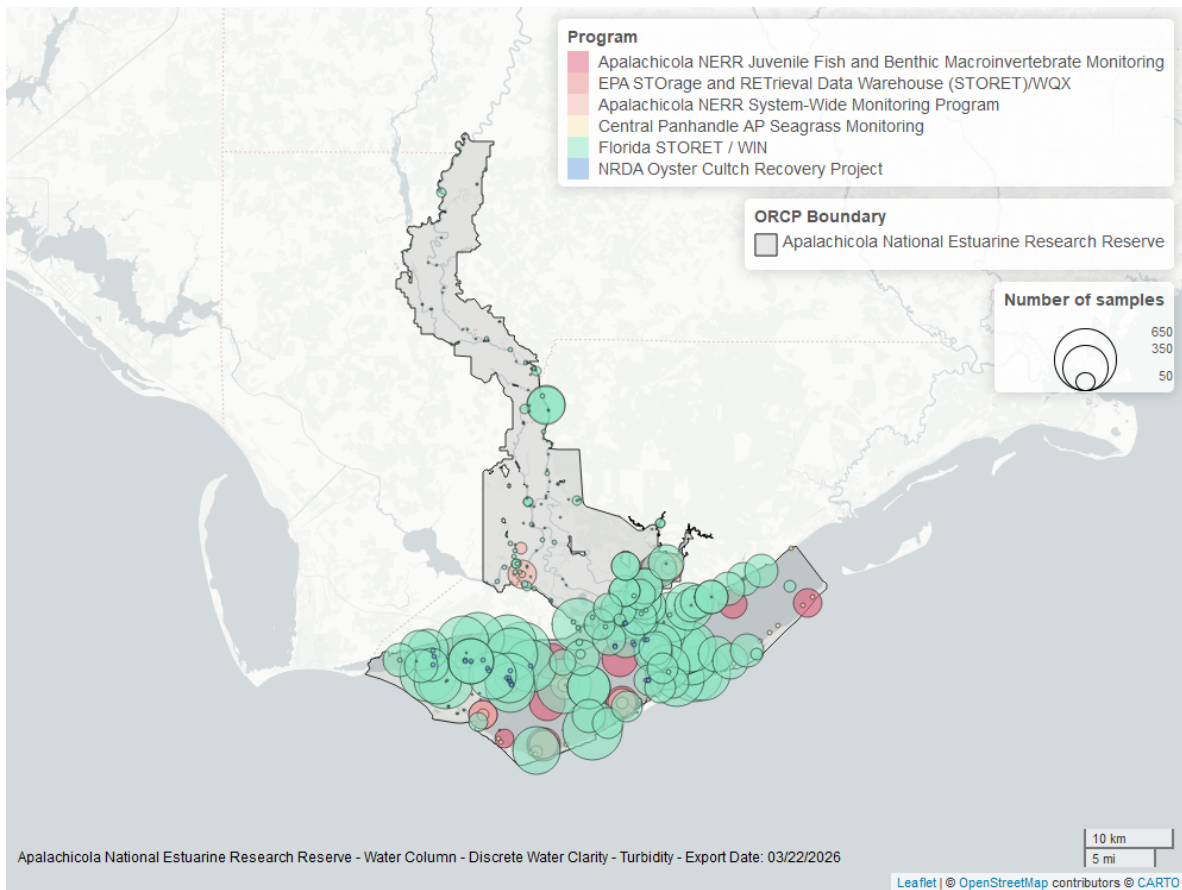


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Total Suspended Solids - Discrete

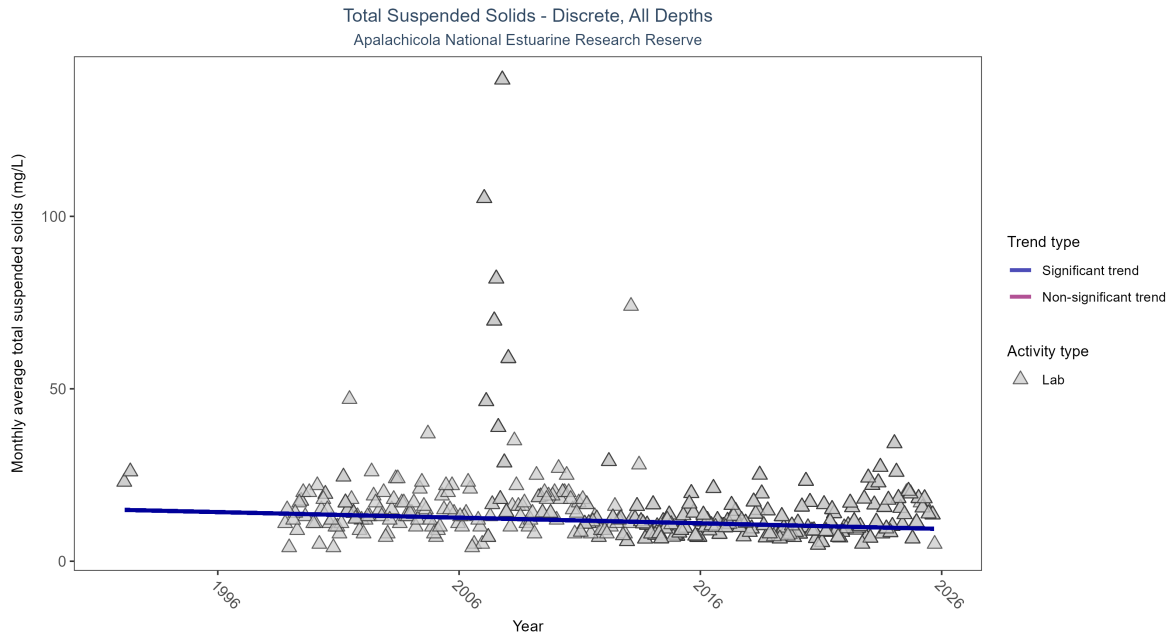


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly decreasing trend	3480	29	1992 - 2025	10	-0.1794	14.90129	-0.1625	0

Monthly average total suspended solids decreased by 0.16 mg/L per year, indicating an increase in water clarity.

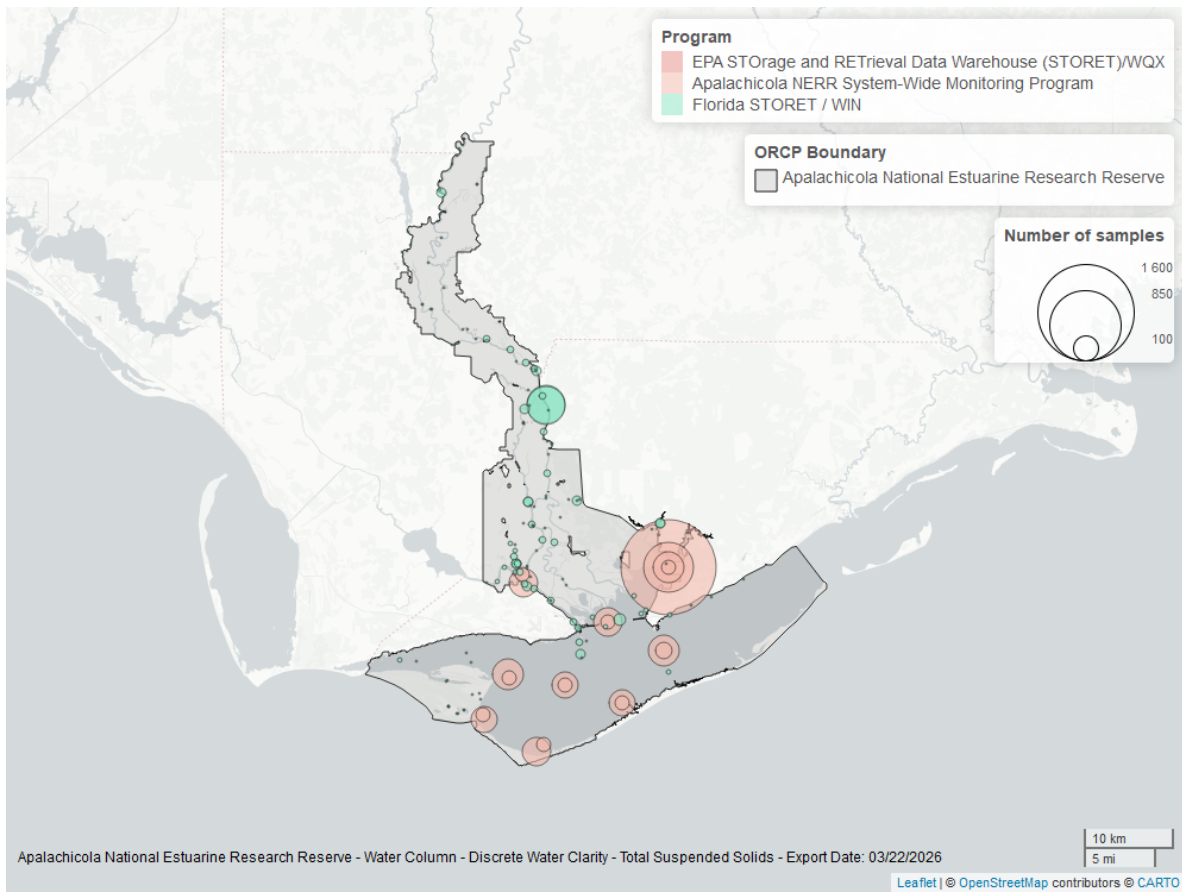


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Uncorrected for Pheophytin - Discrete

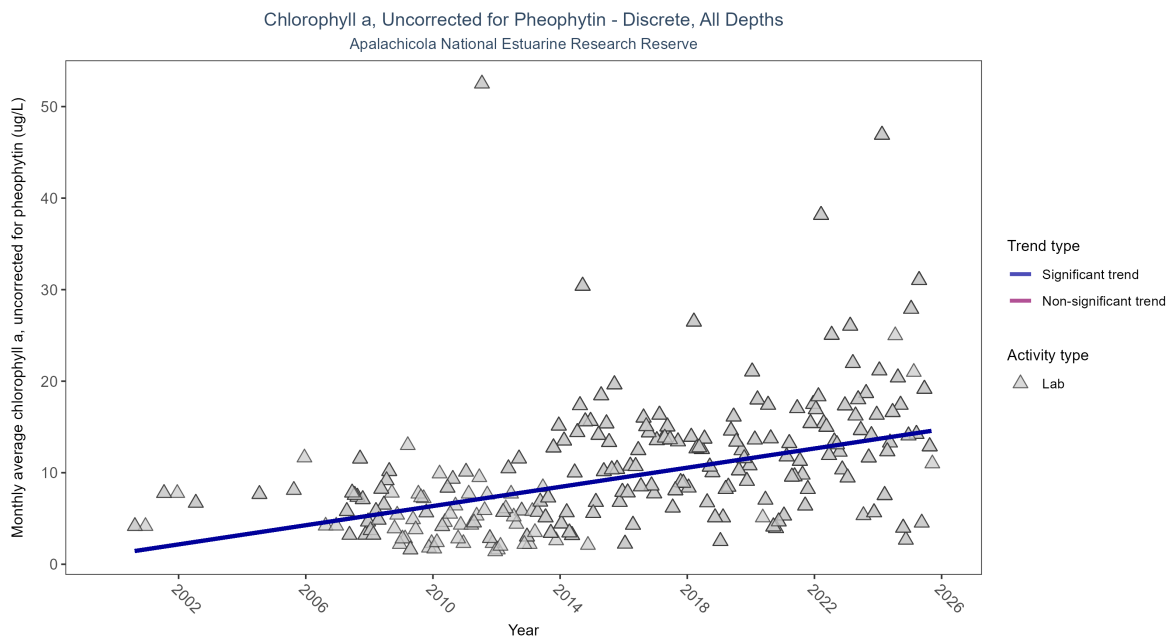


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	3216	25	2000 - 2025	10	0.41262	1.12156	0.52381	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.52 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

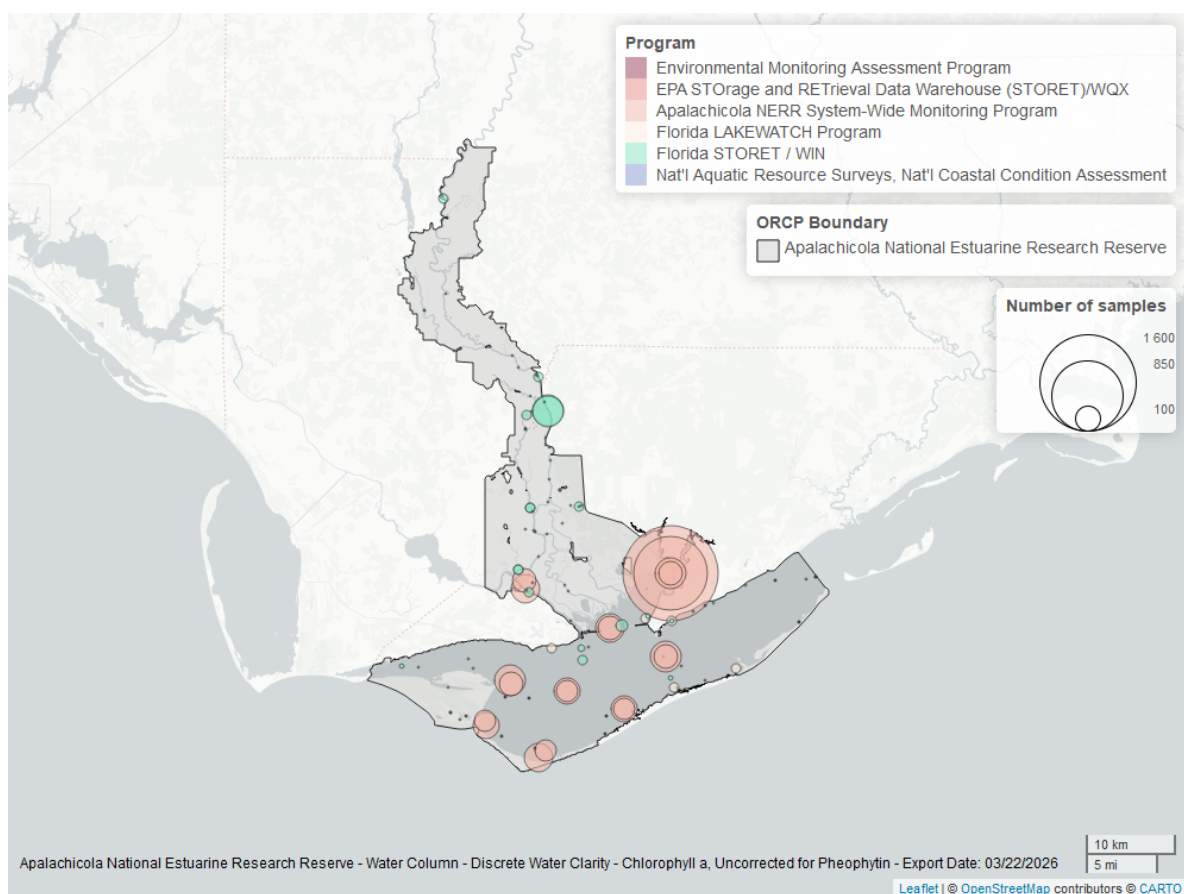


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Chlorophyll a, Corrected for Pheophytin - Discrete

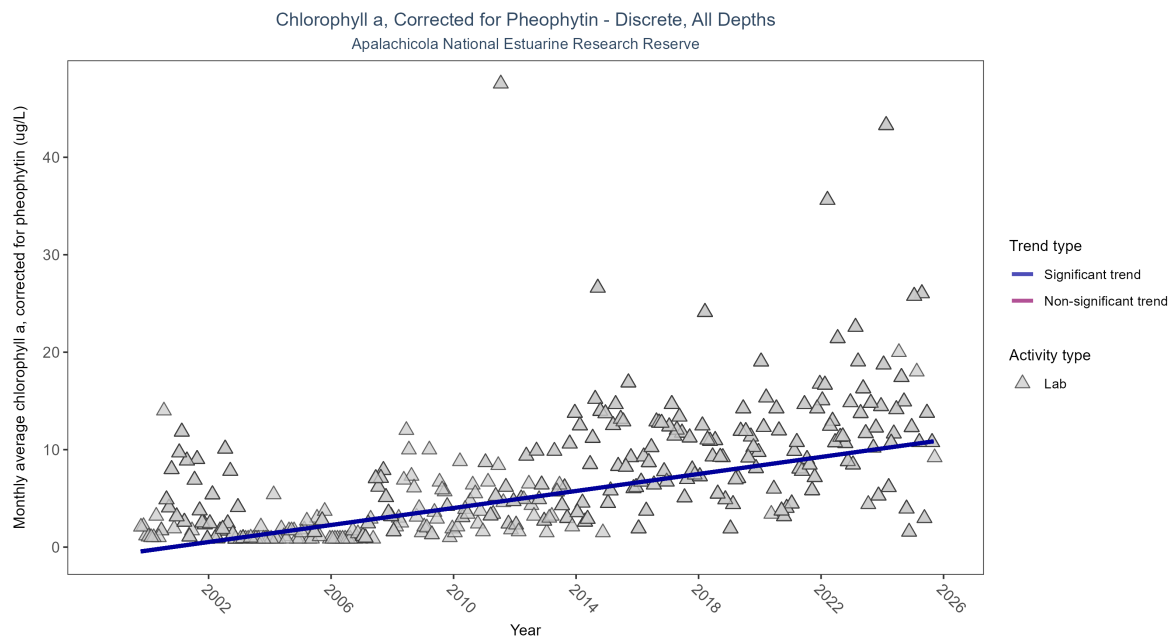


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Significantly increasing trend	3877	27	1999 - 2025	7.5	0.51683	-0.78714	0.43636	0

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.44 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

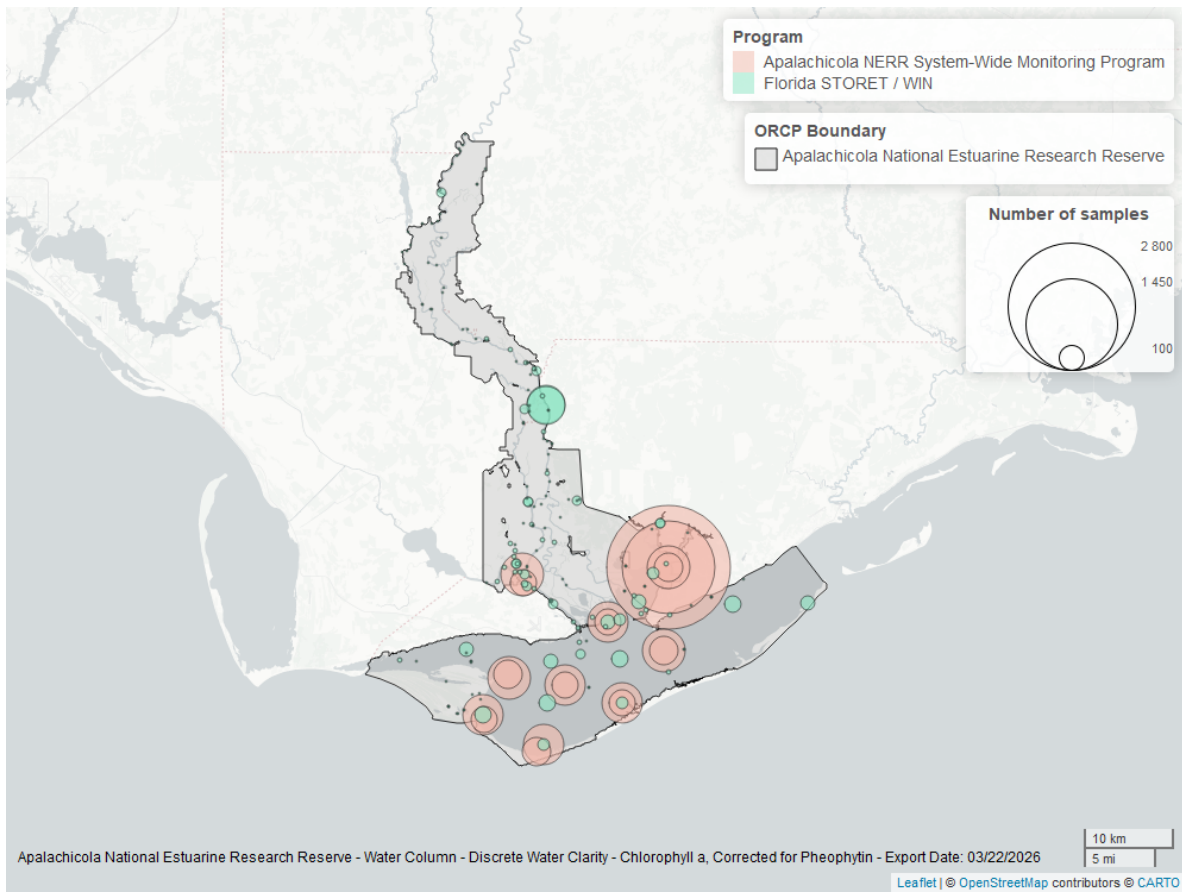


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Secchi Depth - Discrete

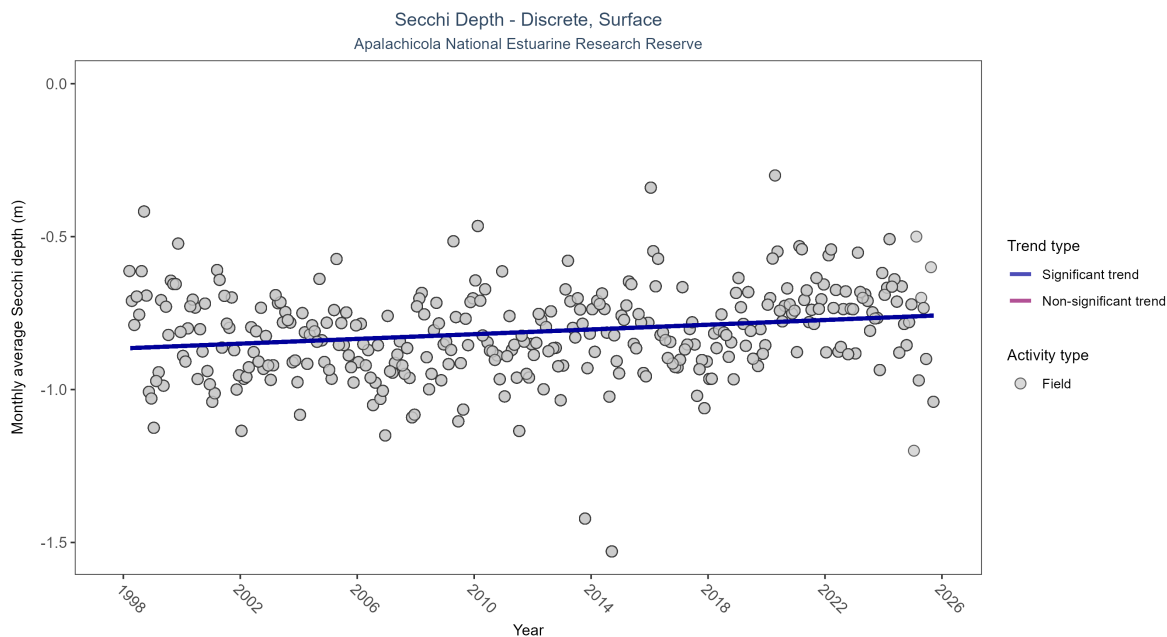


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Significantly increasing trend	34757	28	1998 - 2025	-0.7	0.18004	-0.8654	0.00387	0

Monthly average Secchi depth became shallower by less than 0.01 m per year, indicating a decrease in water clarity.

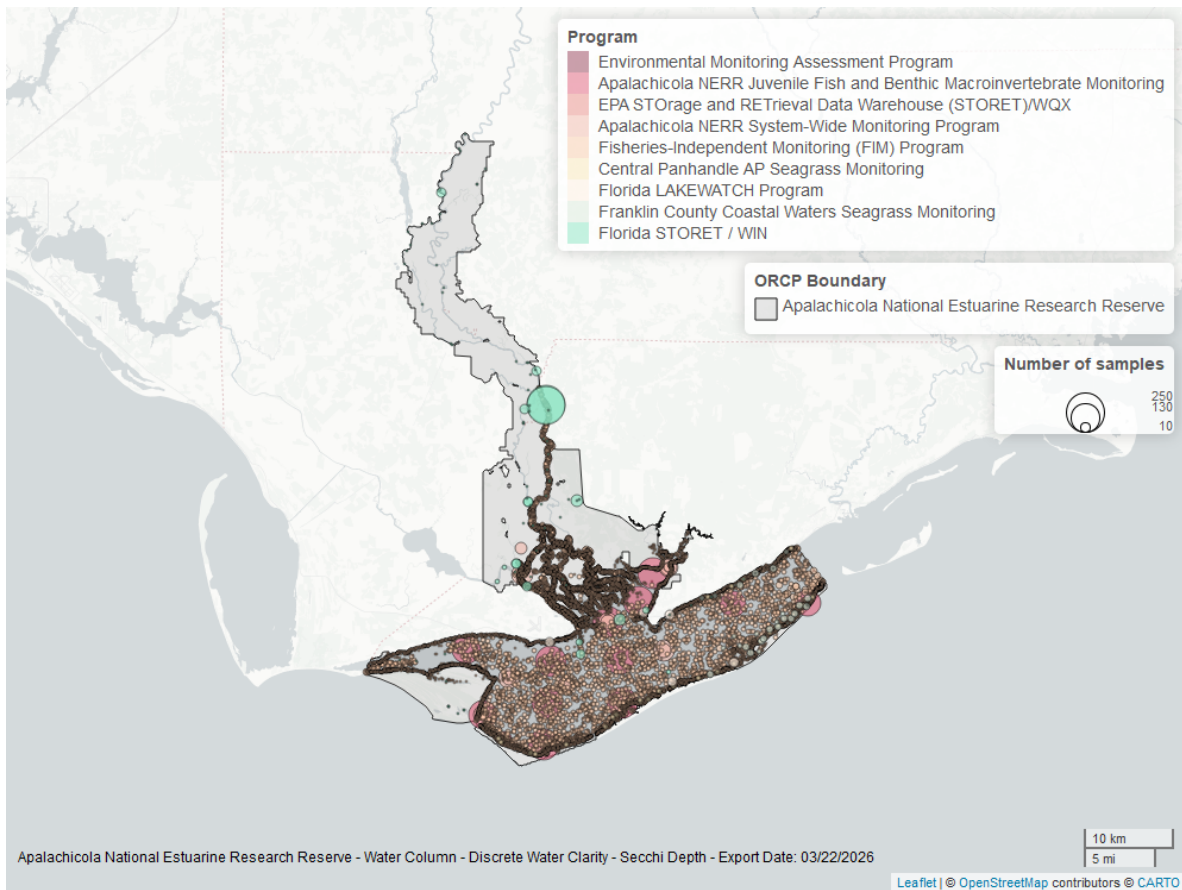


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Colored Dissolved Organic Matter - Discrete

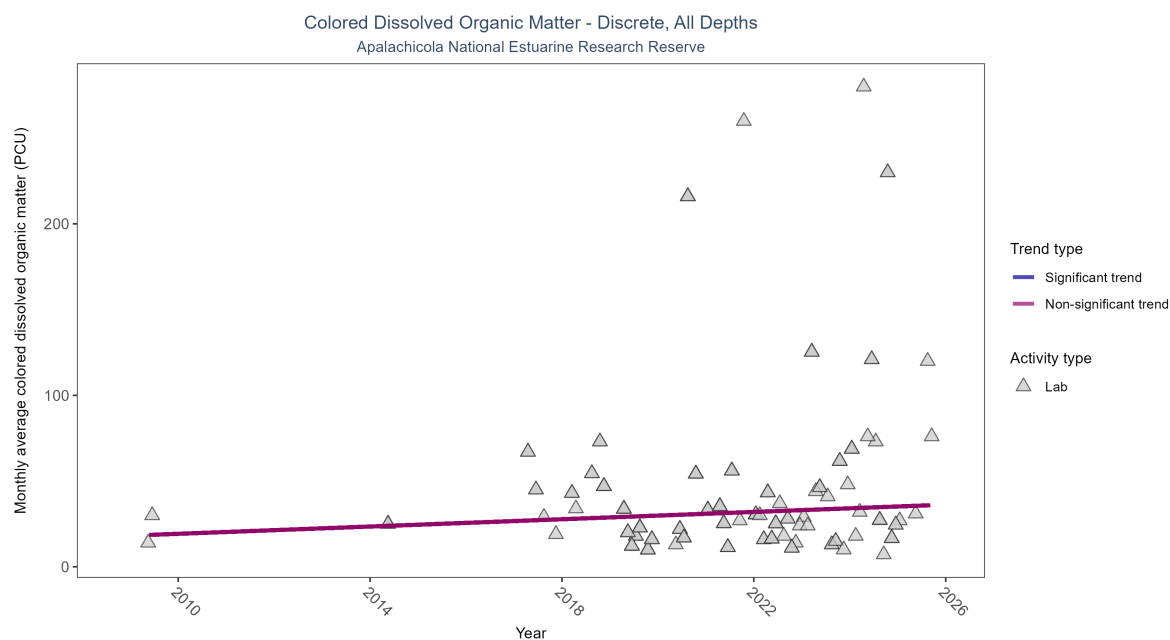


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

Activity Type	Statistical Trend	Sample Count	Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No significant trend	157	11	2009 - 2025	25	0.08184	18.16667	1.0625	0.1916

Colored dissolved organic matter showed no detectable trend between 2009 and 2025.

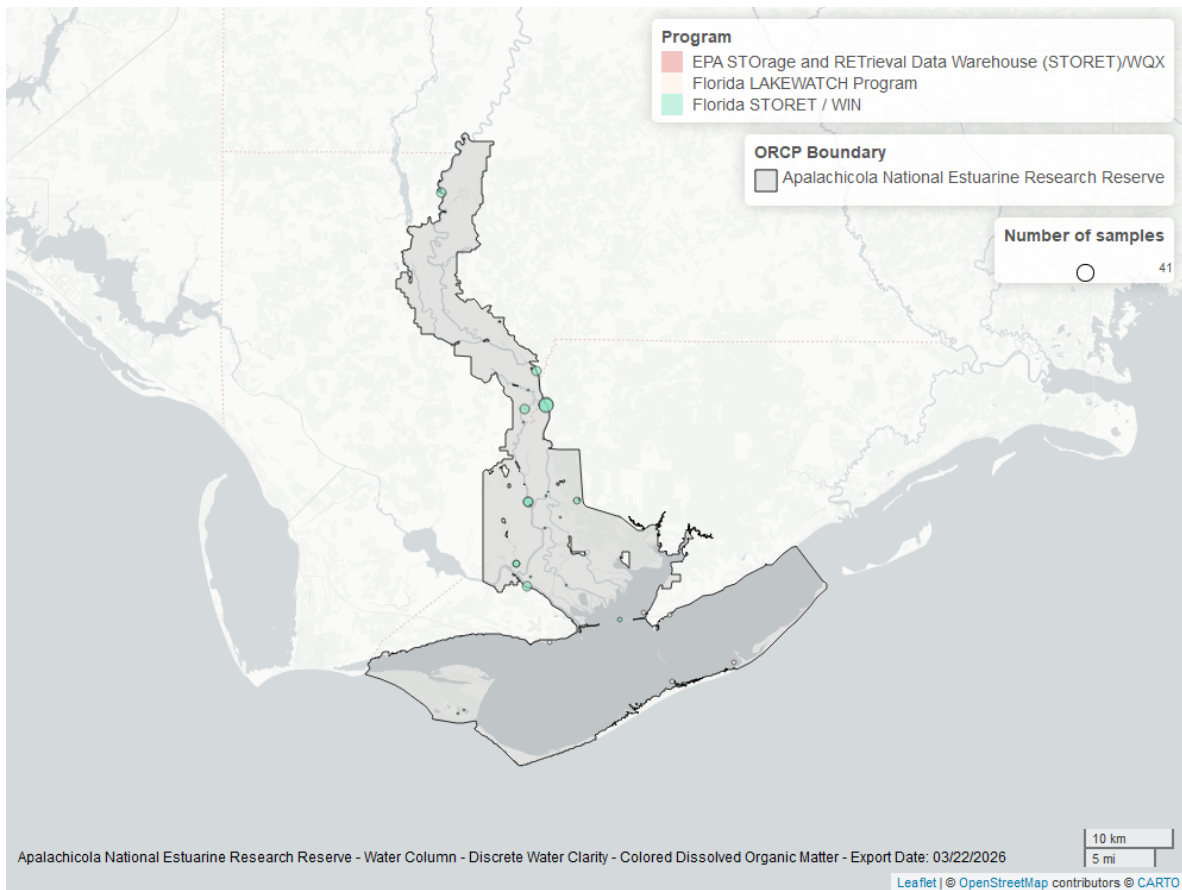


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *Apalachicola National Estuarine Research Reserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.